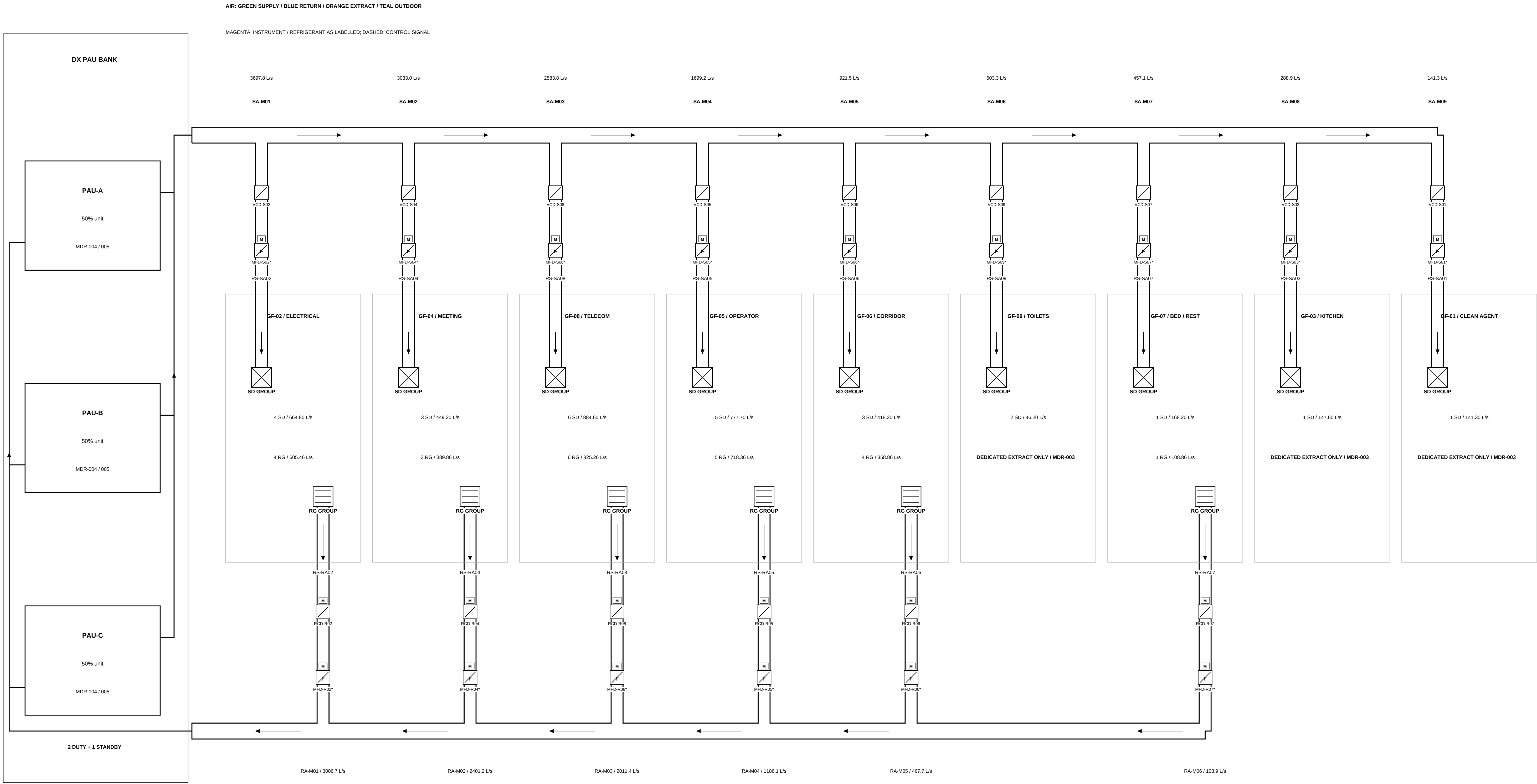


MAIN DUCT AND RETURN - WHOLE SYSTEM AIRFLOW METHOD



HOW THE METHOD WORKS

DX cooling takes place inside the PAUs. Refrigerant stays within the selected factory DX system; supply and return mains carry AIR. Common SUPPLY and RETURN mains stay on the ROOF. Nine supply drops and six return risers connect room branches to the shared headers; every PAU connects to the same header bank. Terminal groups and all instruments are expanded on the following sheets.

Supply 3697.80 L/s; assumed return 3006.68 L/s; minimum mass-balance makeup 691.12 L/s. DX cooling capacity / outside air / actual pressures remain unselected. Rooms with dedicated extract have no common return branch.

NTS functional air diagram: room order and sequential flow steps are illustrative; actual grouped roof takeoffs / coordinates on MDR-012. MFD* conditional on approved fire boundaries; RCD pressure modulation is a proposal. All room flows retained. Main sizes: MDR-011.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

WHOLE SYSTEM AIRFLOW METHOD

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-001

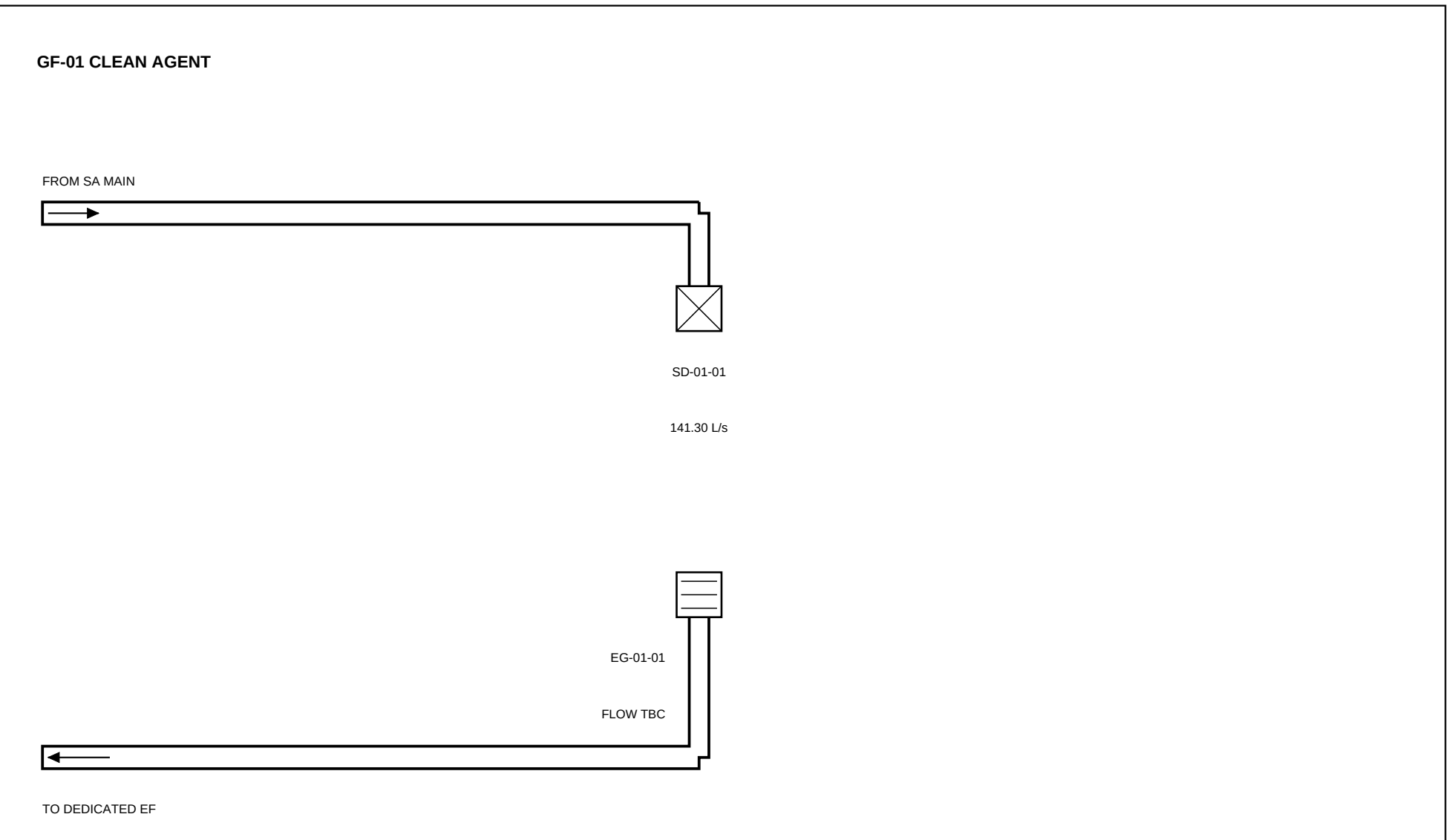
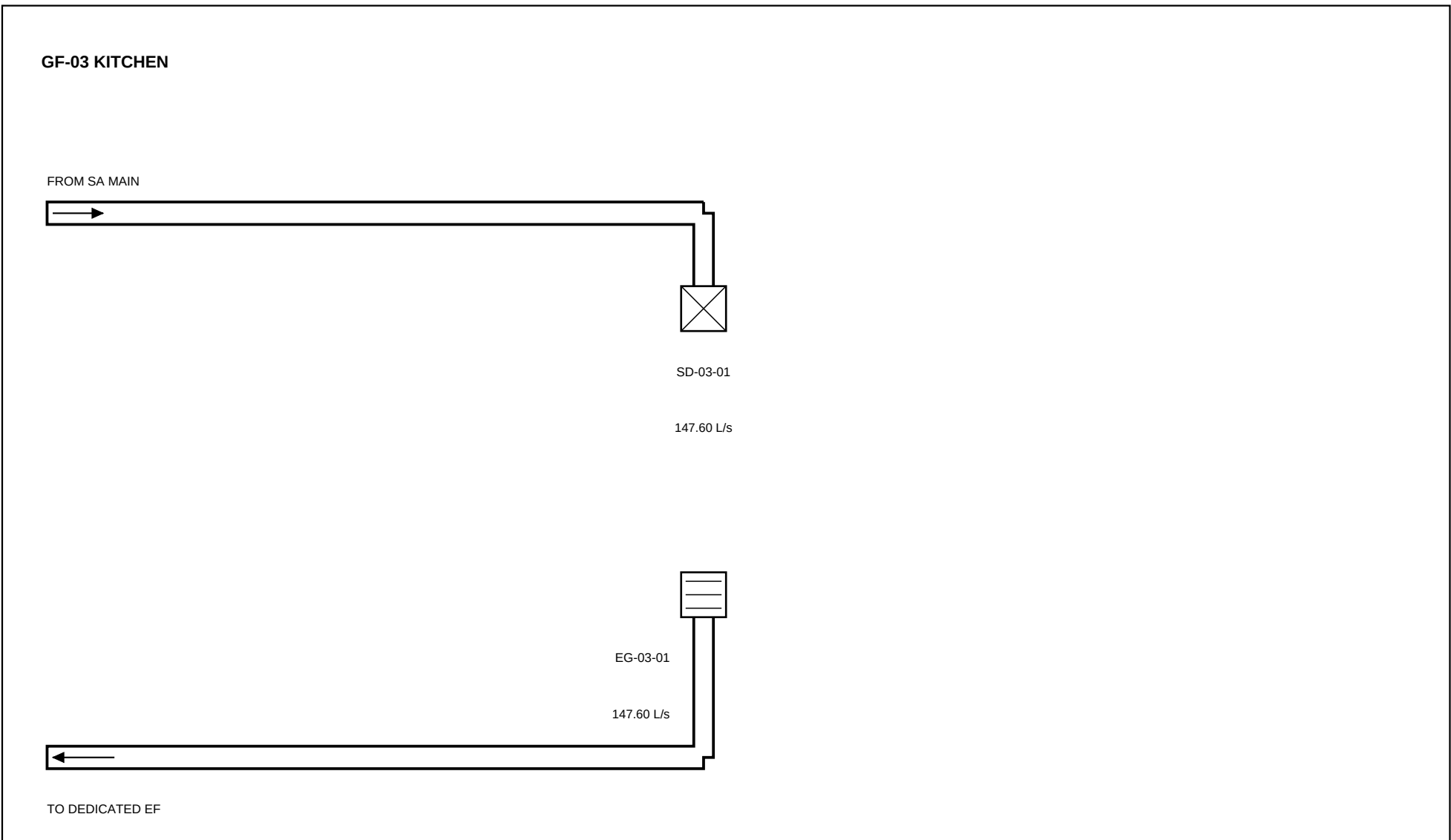
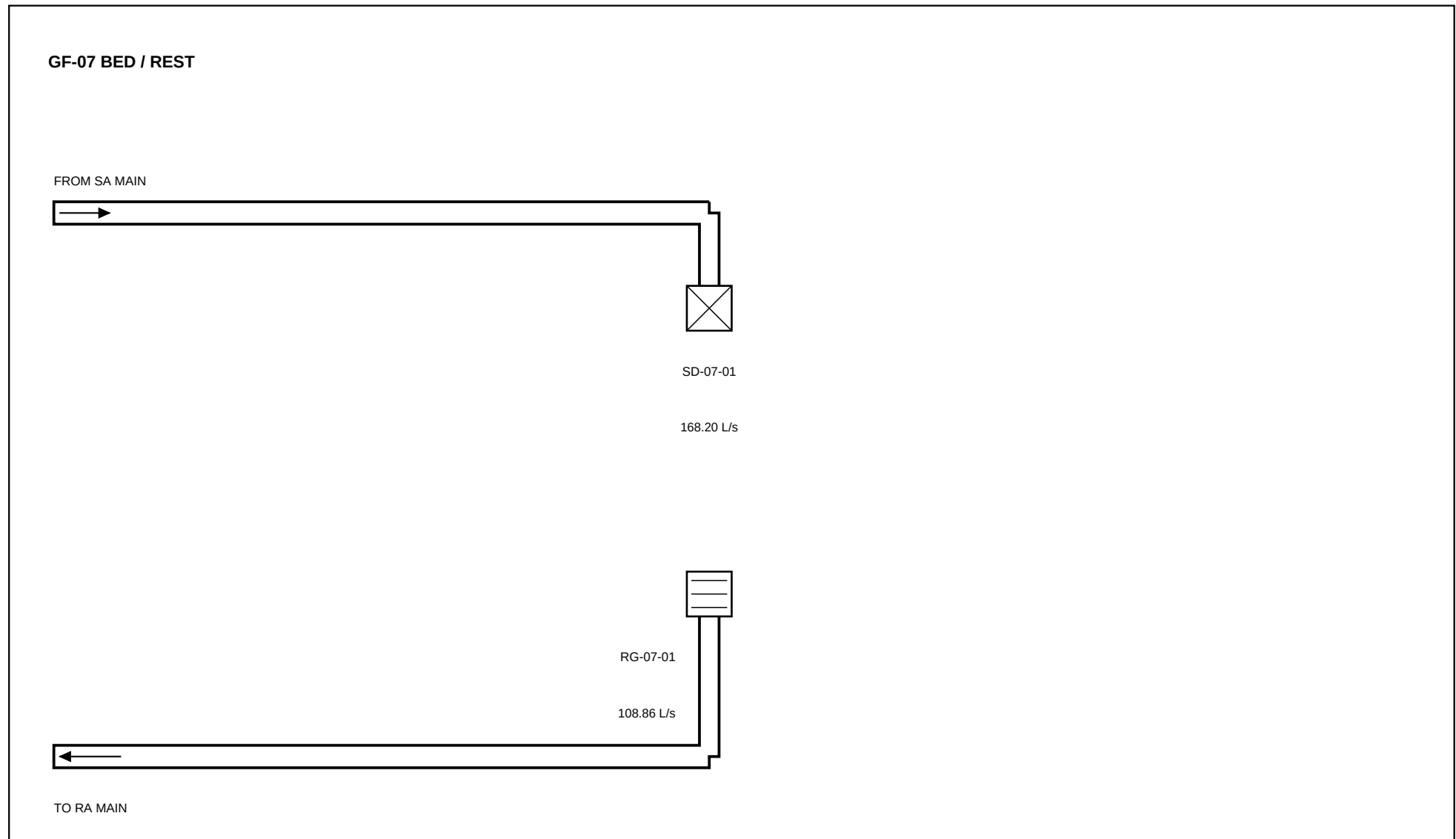
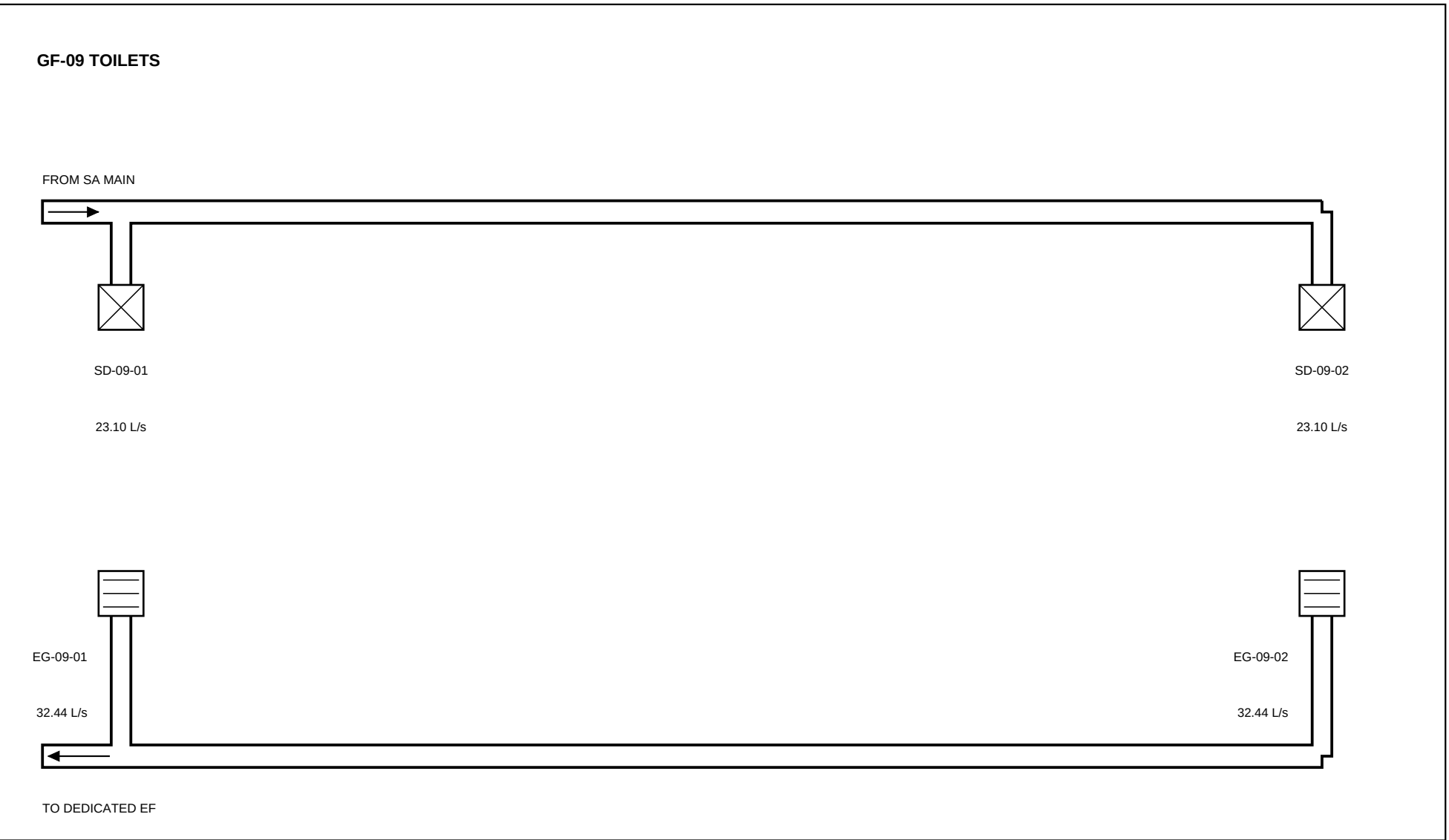
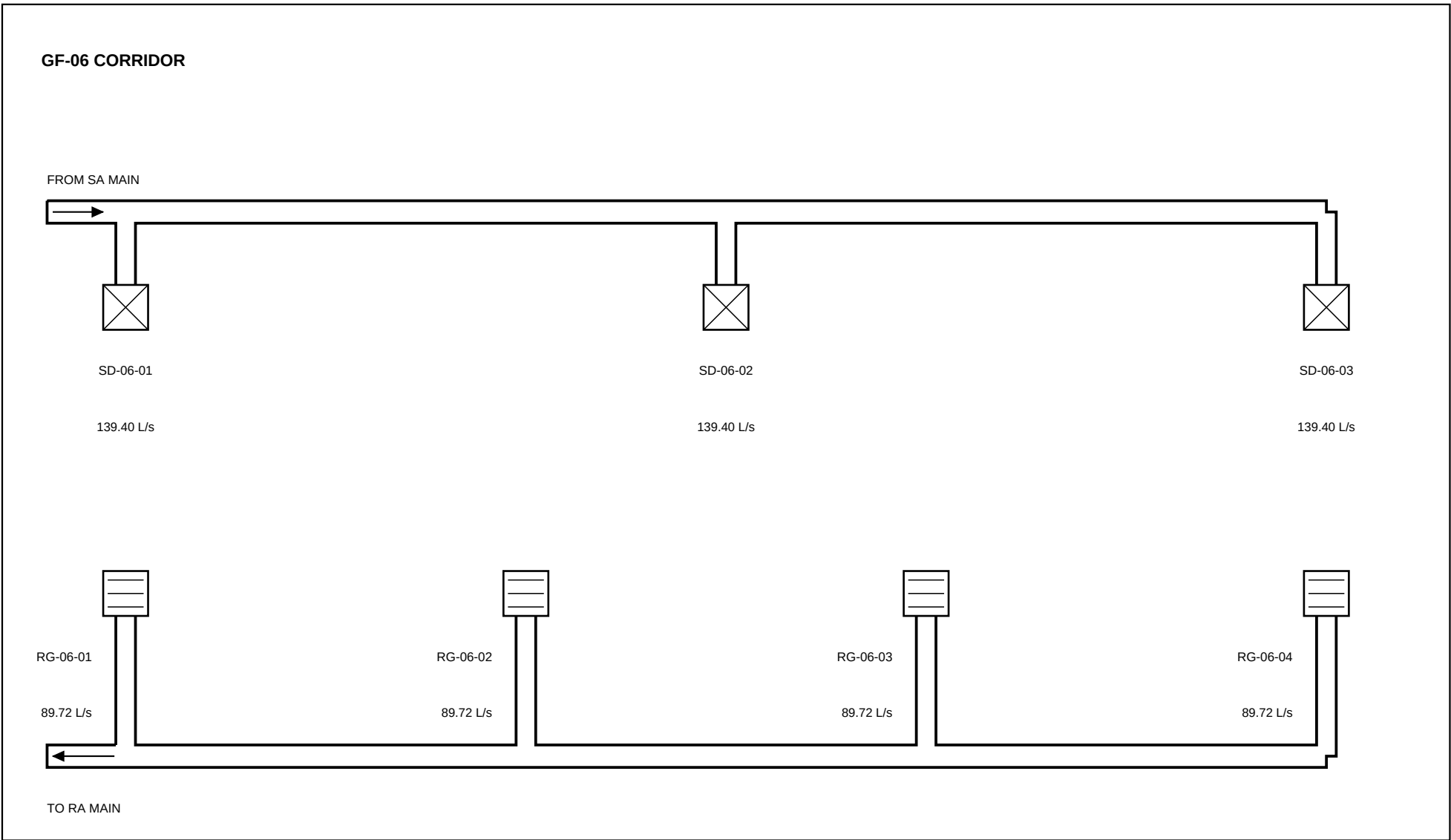
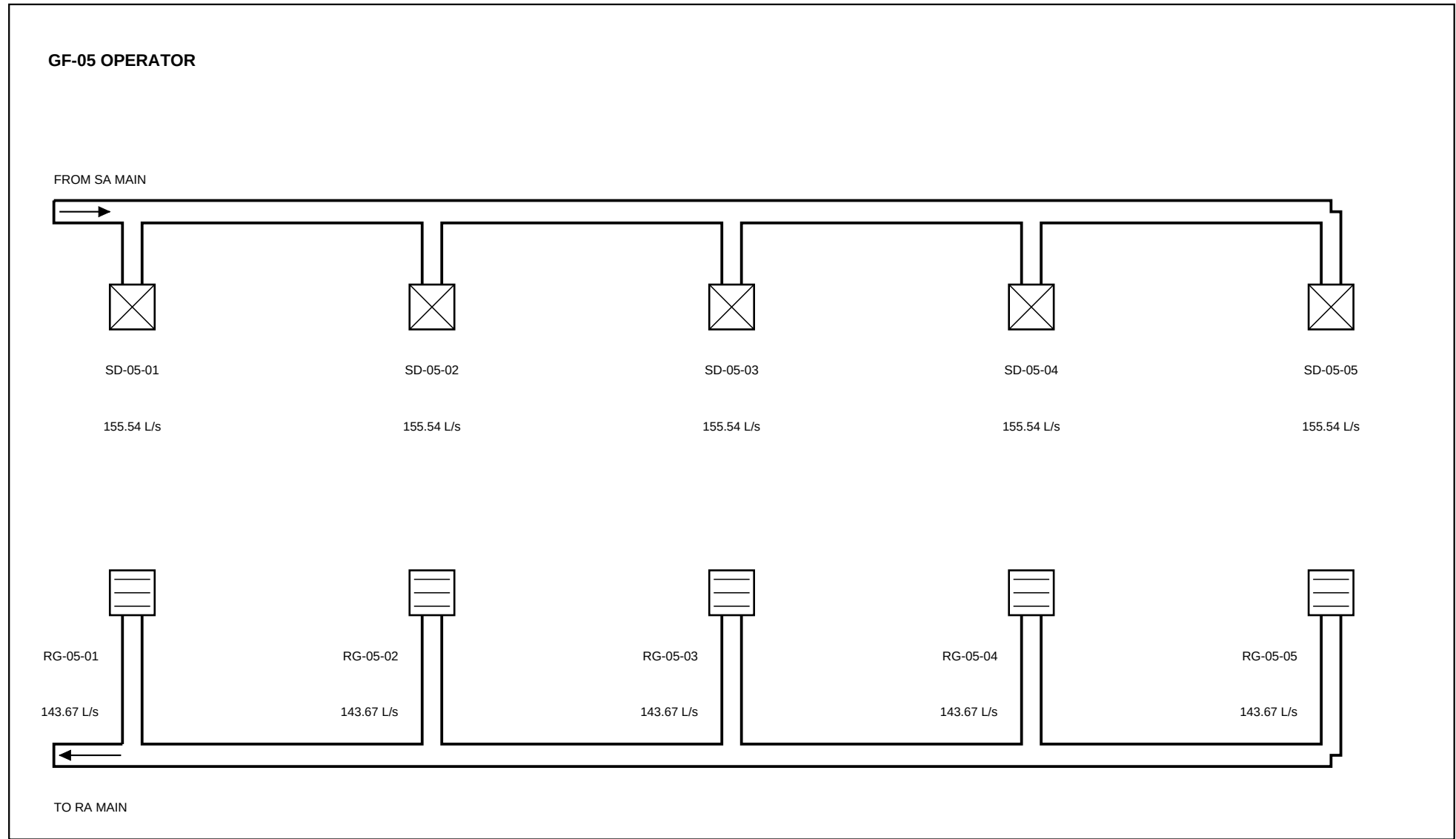
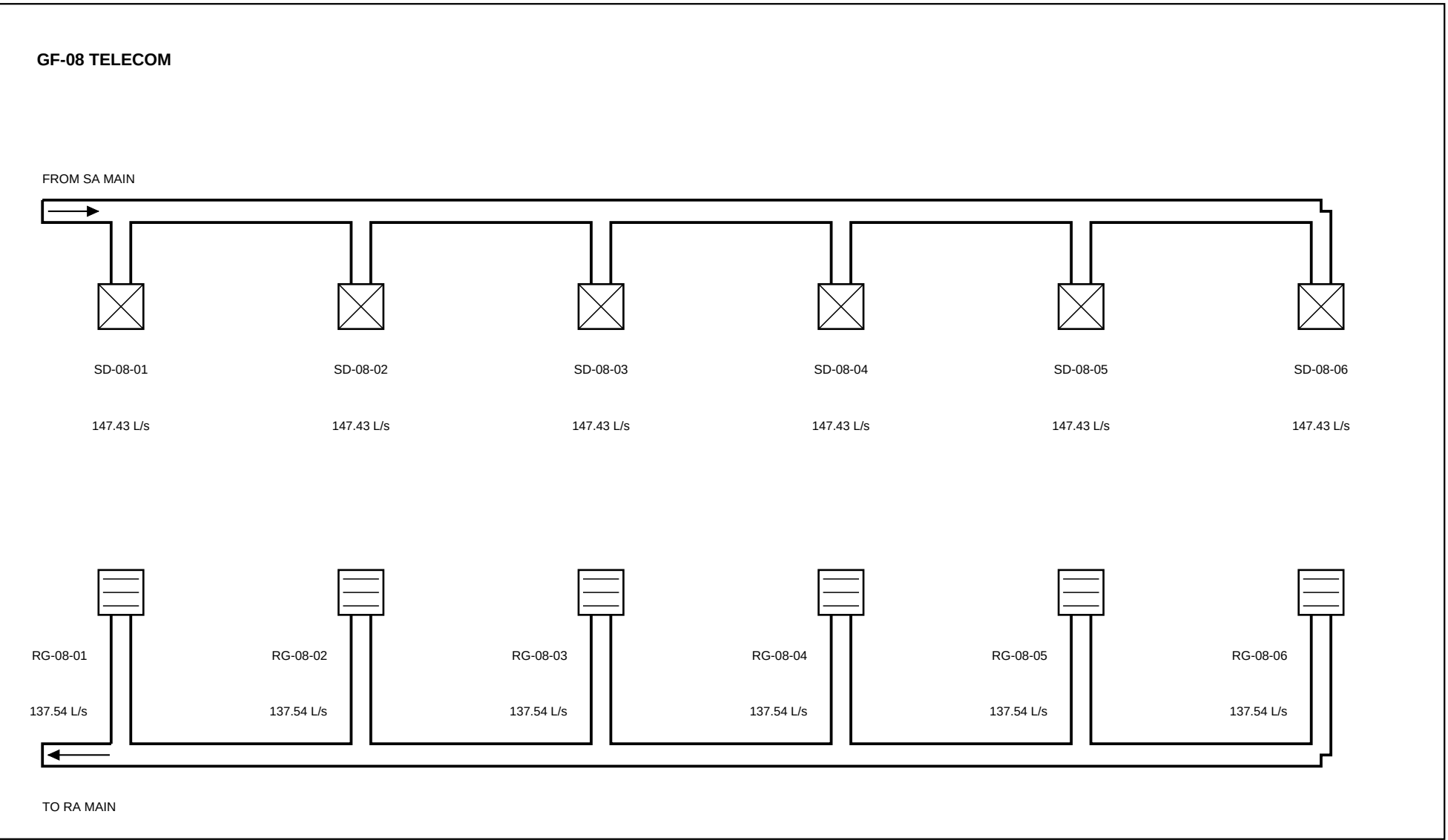
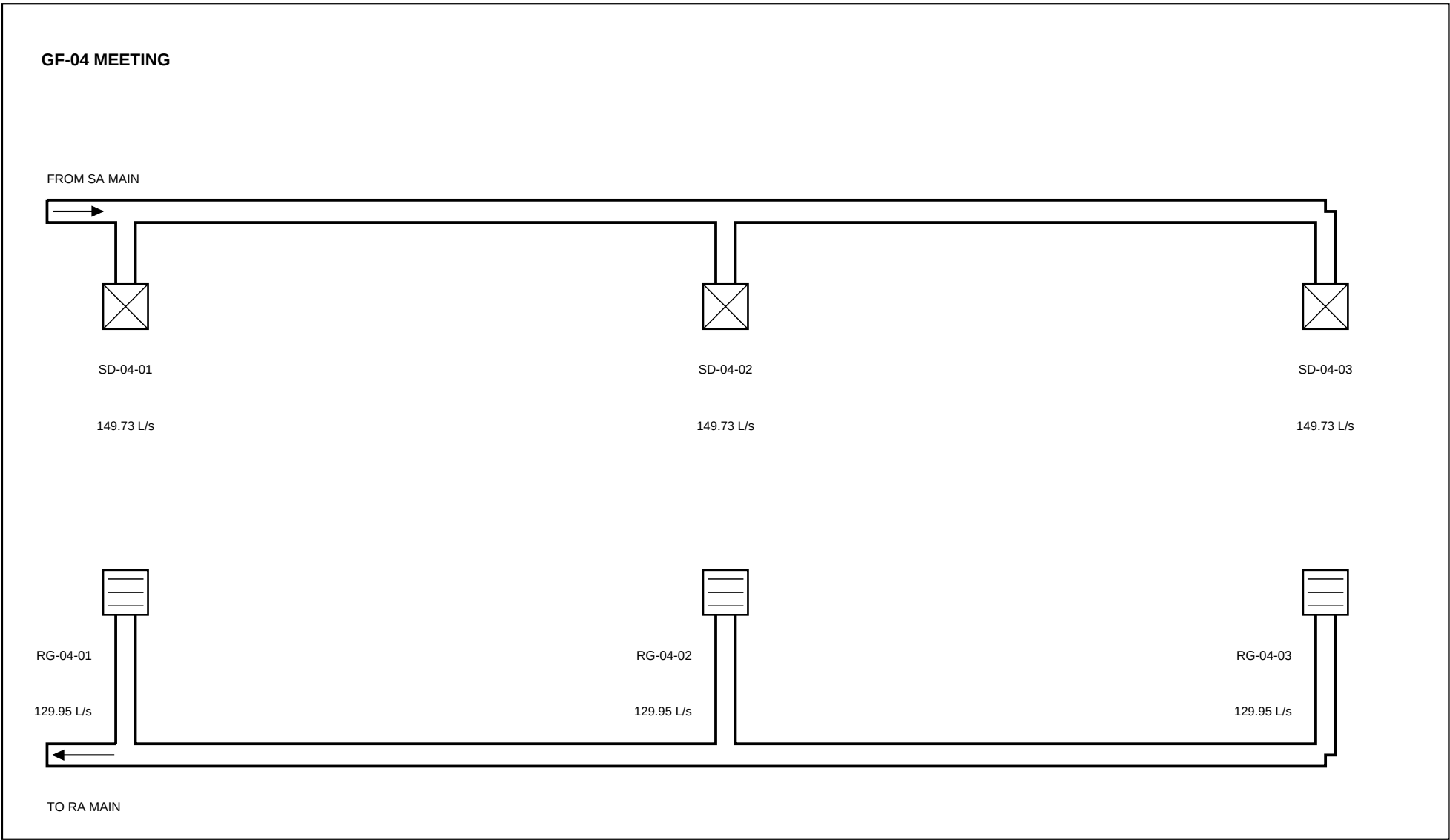
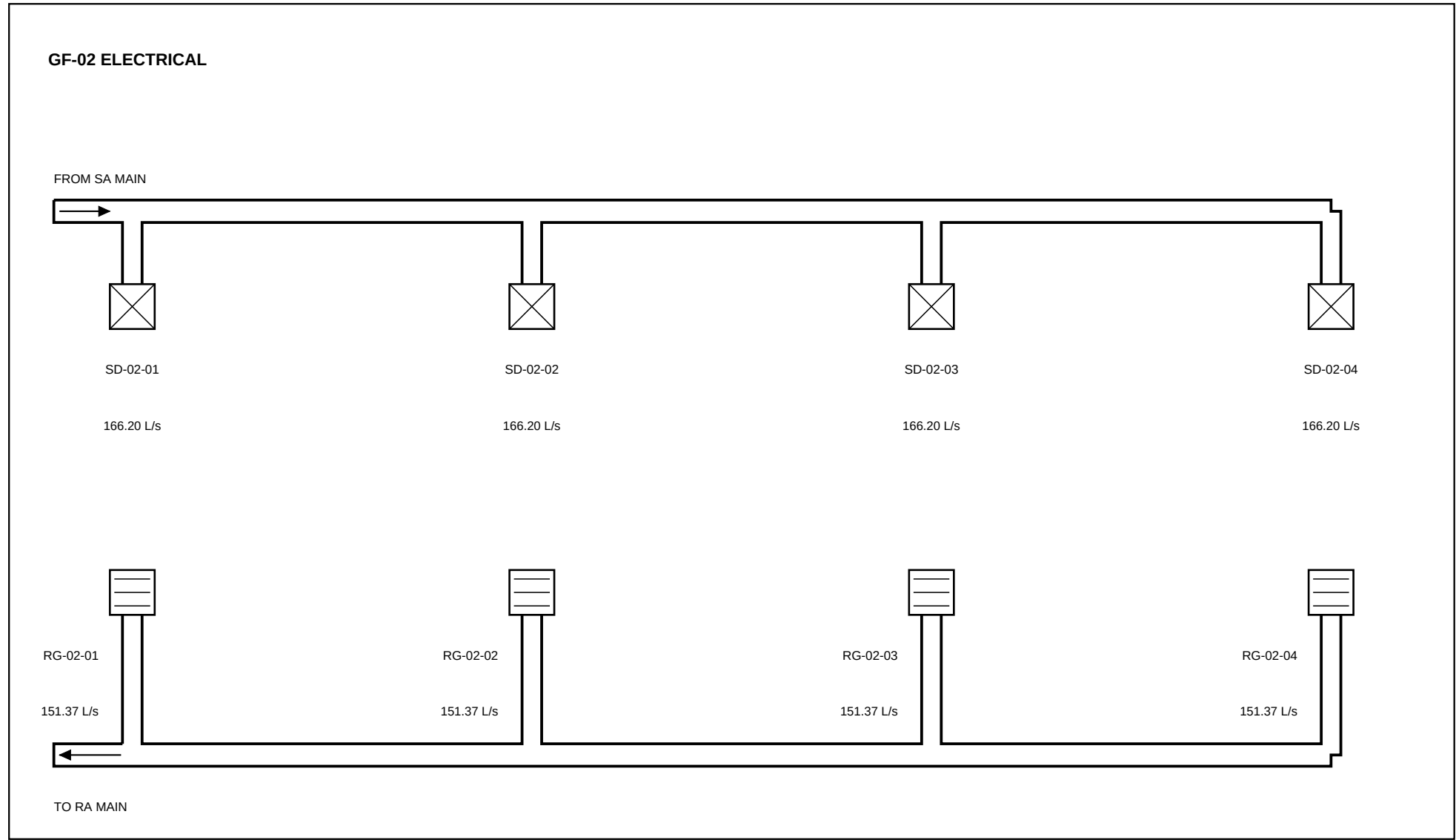
SCALE: NTS AT A1

METHOD REVISION 01

08 OCT 2026

SHEET 1 OF 14

MAIN DUCT AND RETURN - ALL ROOM DIFFUSERS, RETURN GRILLES AND EXTRACT PICKUPS



All S3 terminal tags / duties are retained. Glyph represents the registered terminal face; this sheet is a functional room header diagram, not ceiling setting-out. Terminal neck / face sizes and coordinates retained in the workbook.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

ALL ROOM DIFFUSERS, RETURN GRILLES AND EXTRACT PICKUPS

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-002

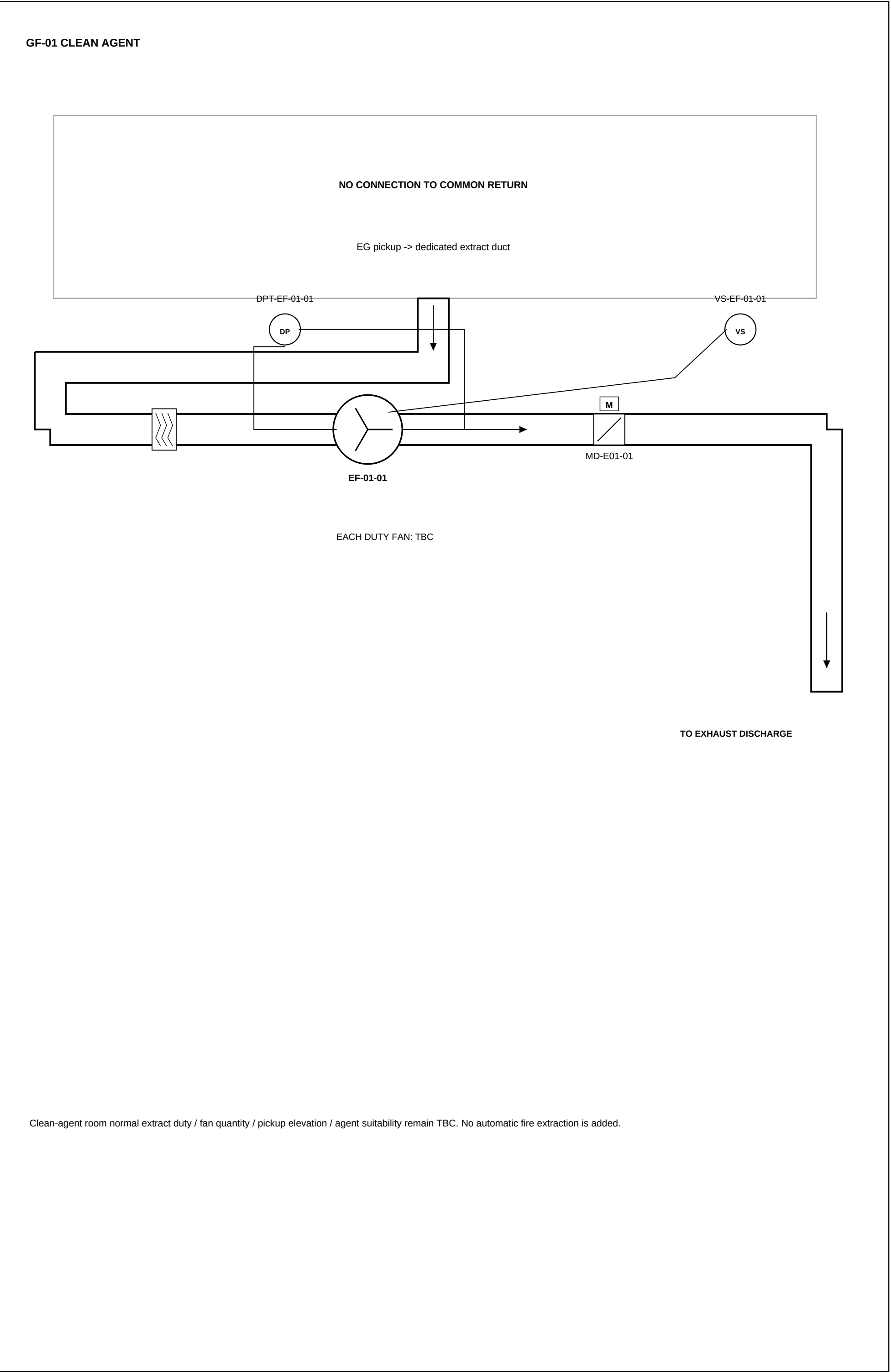
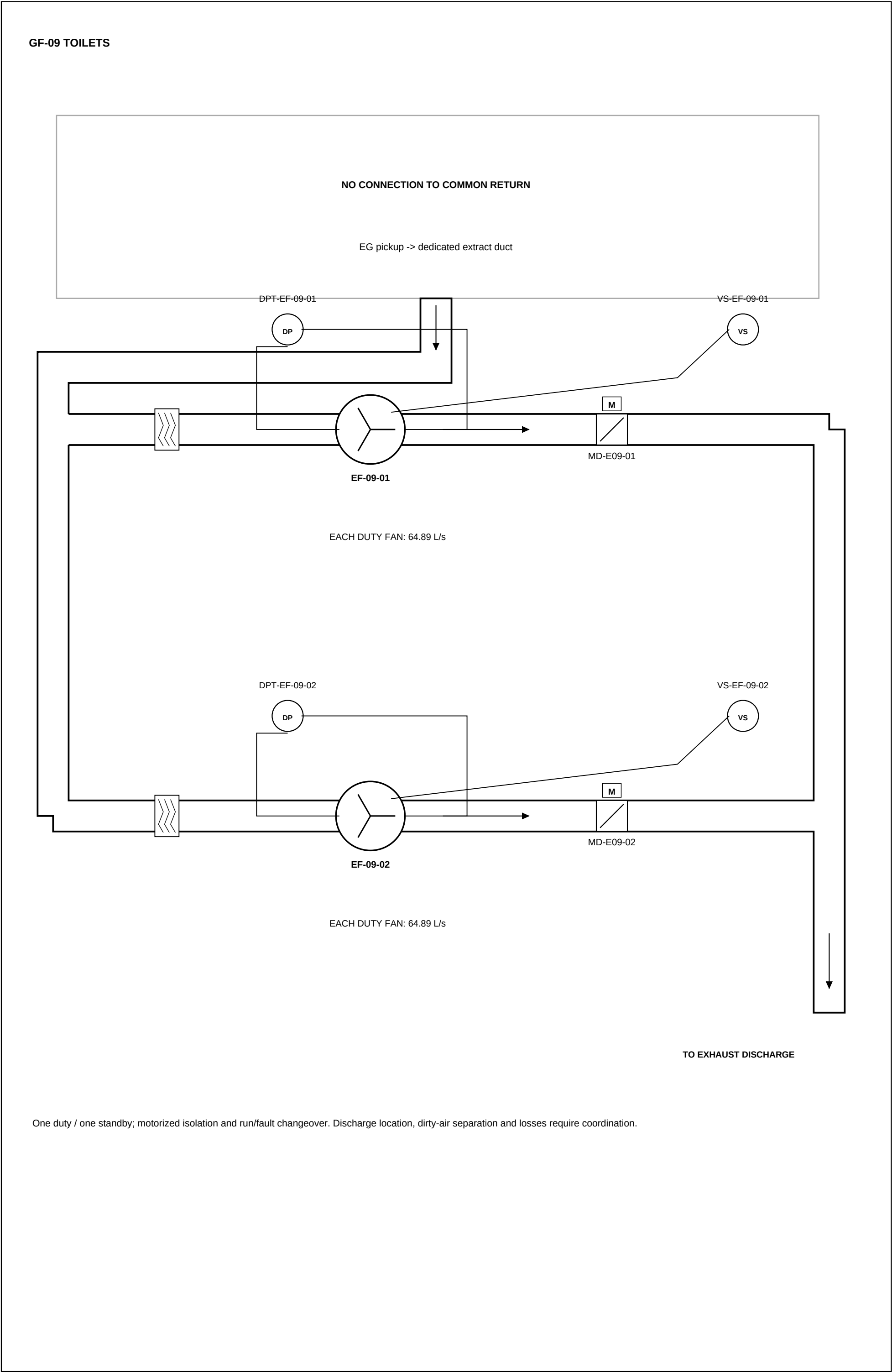
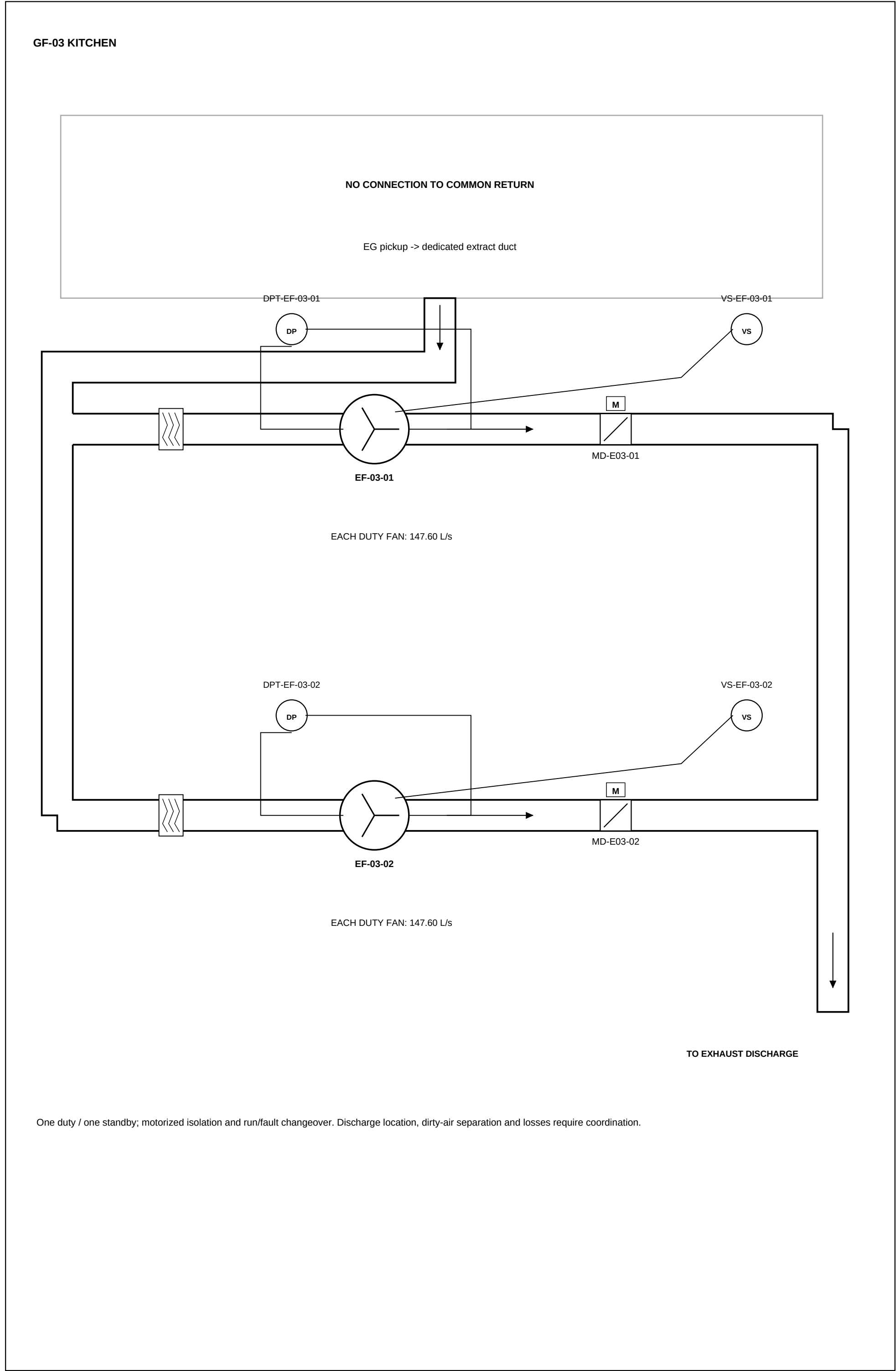
SCALE: NTS AT A1

METHOD REVISION 01

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SHEET 2 OF 14

MAIN DUCT AND RETURN - DEDICATED EXTRACTS AND RETURN AIR SEPARATION



AIR BALANCE / FIRE REQUIREMENTS

Toilet SA 46.20 minus EA 64.89 = -18.69 L/s; kitchen SA 147.60 minus EA 147.60 = zero. These do not establish the previous +25 Pa targets. Agree transfer / hygiene / pressure strategy using measured leakage and project requirements.

Confirmed fire stops every HVAC fan and heater; fire reset is manual. No automatic smoke extraction or post-fire purge is implied. Dedicated extract fan outlet / discharge duty and CAG selection remain open.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

DEDICATED EXTRACTS AND RETURN AIR SEPARATION

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-003

SCALE: NTS AT A1

METHOD REVISION 01

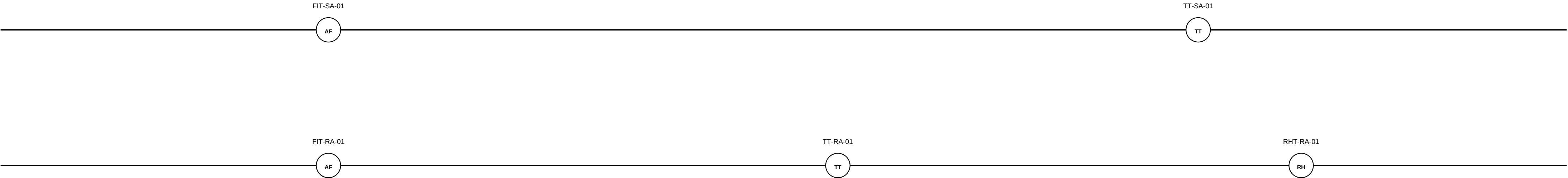
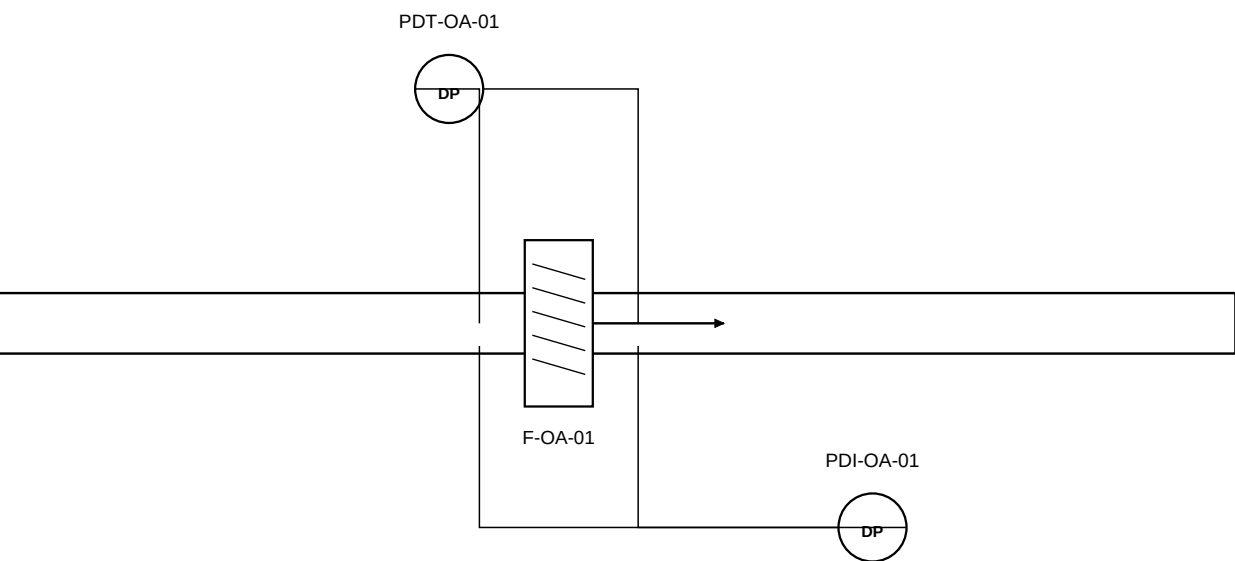
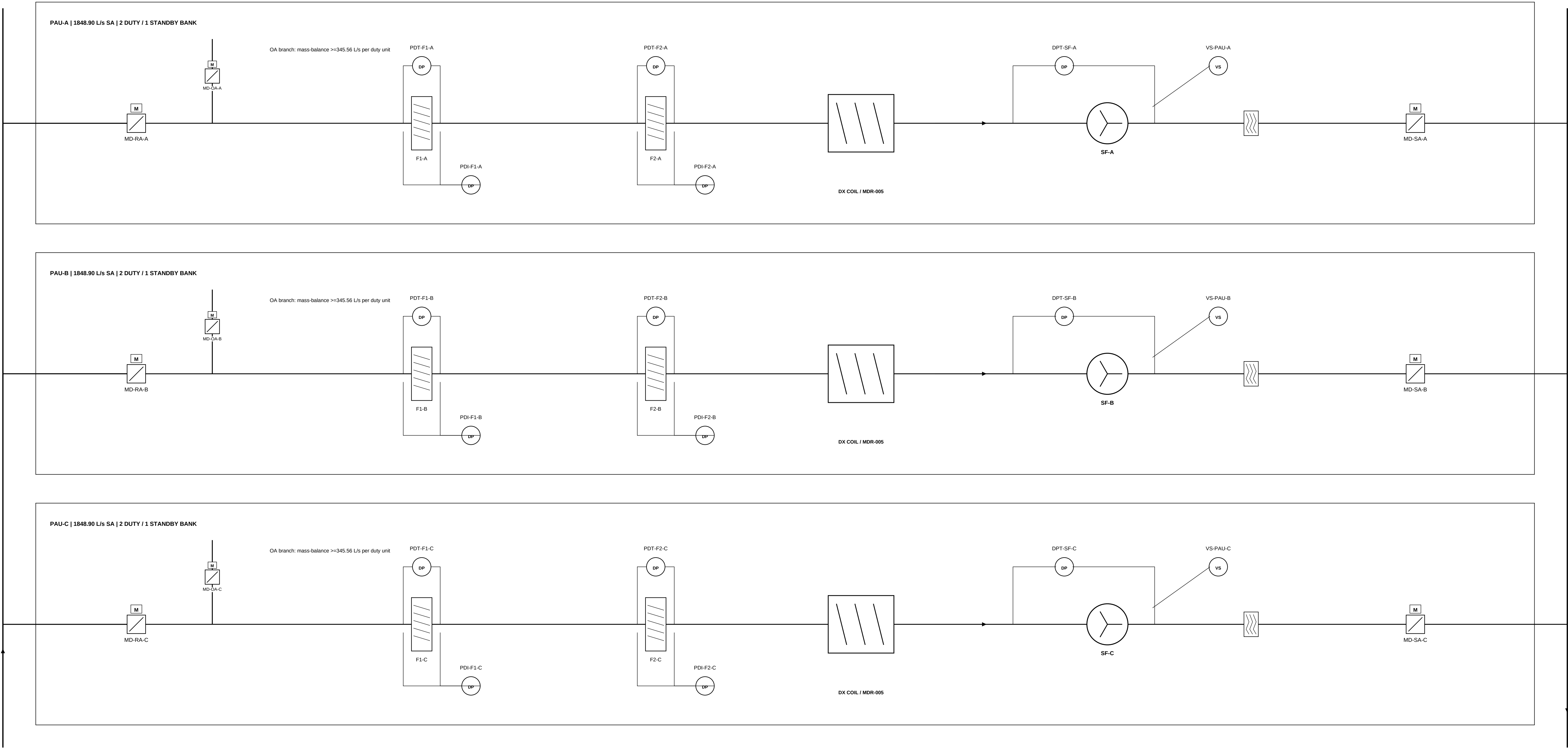
08 OCT 2026

SHEET 3 OF 14

MAIN DUCT AND RETURN - DX PAU BANK, COMMON HEADERS AND AIR-SIDE INSTRUMENTS

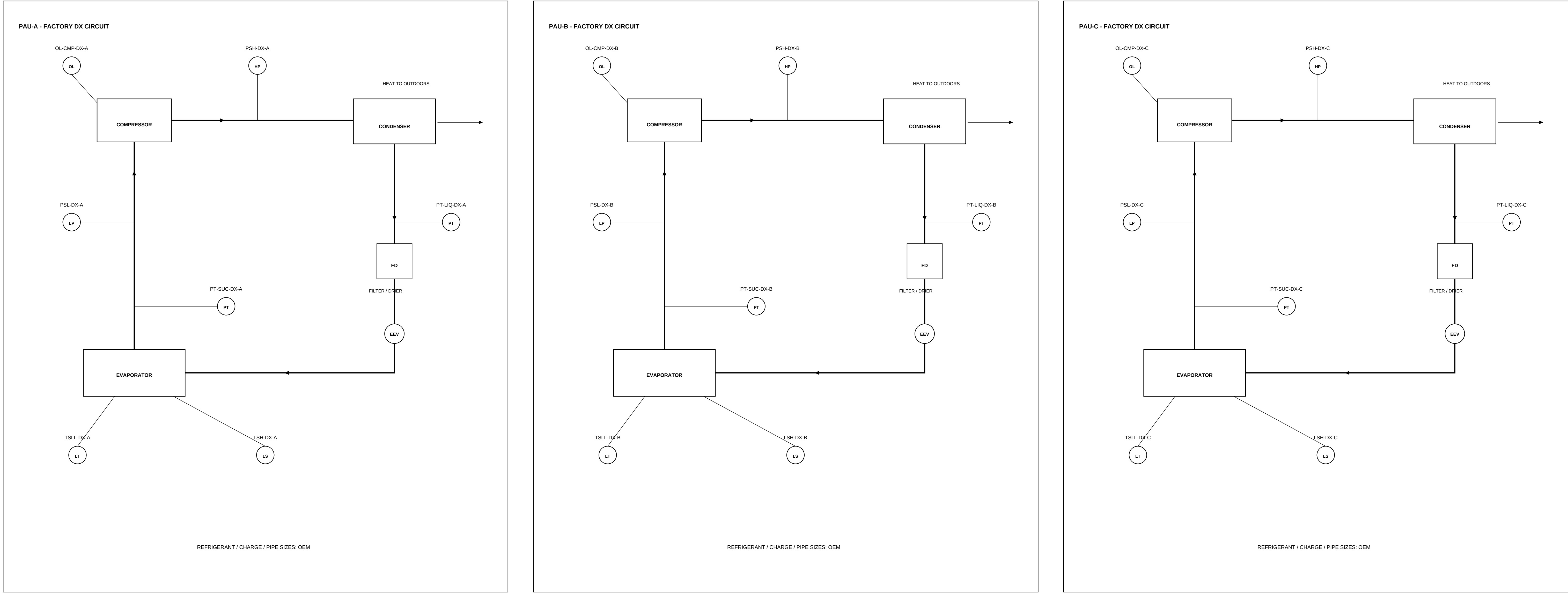
COMMON RA

COMMON SA

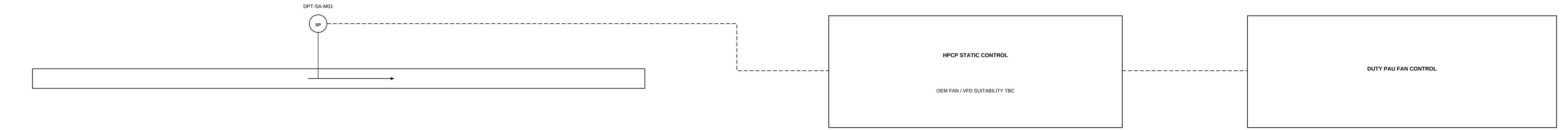


Standby PAU supply / return / OA paths close and prove isolated; duty paths open and prove before start. Common airflow stations require OEM straight runs. Unit nozzle routing, fan ESP, filters, cooling capacity and outdoor air remain unselected.

ADMINISTRATION HVAC / DX PAU OPTION COMMON SUPPLY MAIN + COMMON RETURN MAIN FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION	DRAWING TITLE DX PAU BANK, COMMON HEADERS AND AIR-SIDE INSTRUMENTS ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING	ADM-HVAC-MDR-004 SCALE: NTS AT A1	METHOD REVISION 01 08 OCT 2026 SHEET 4 OF 14
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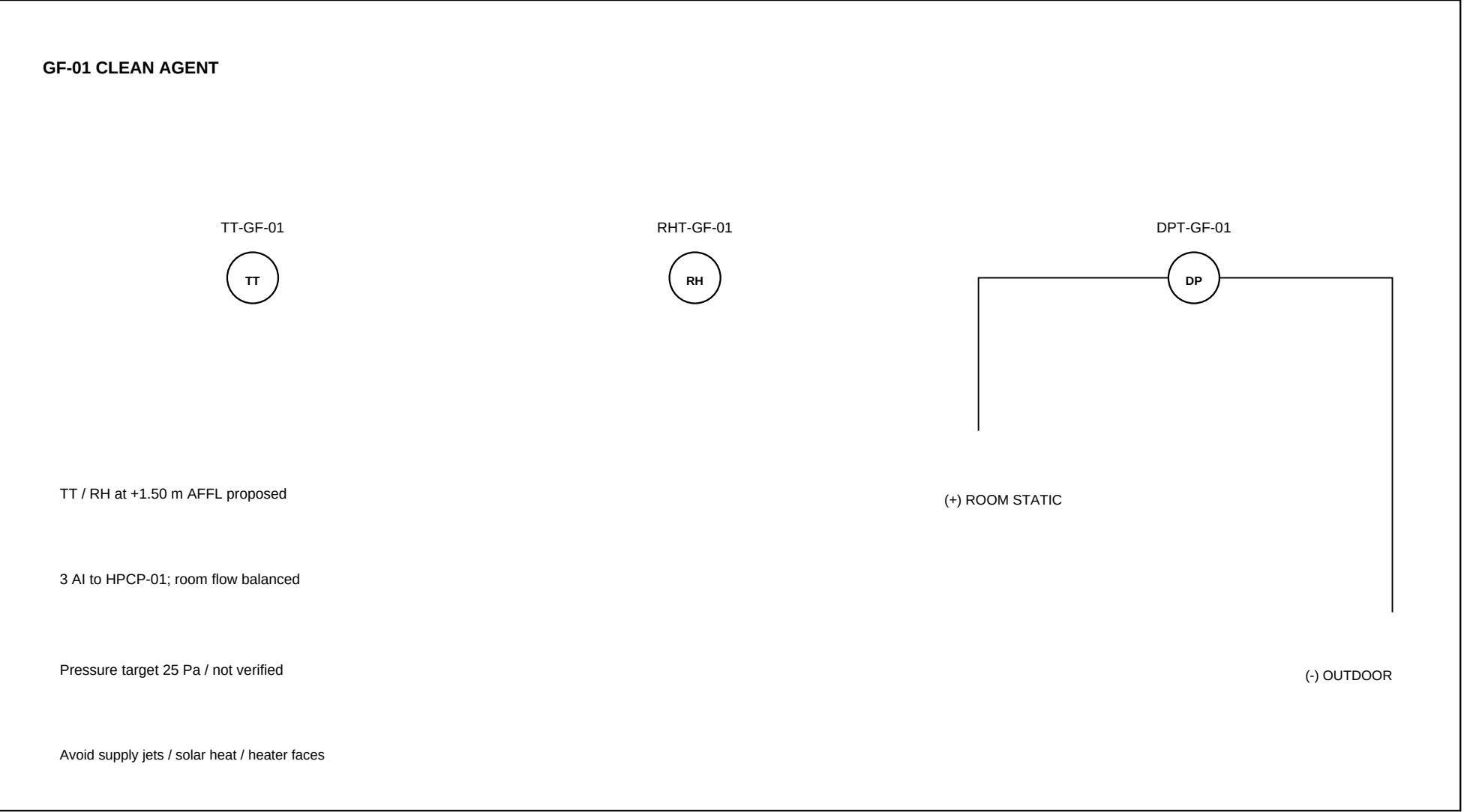
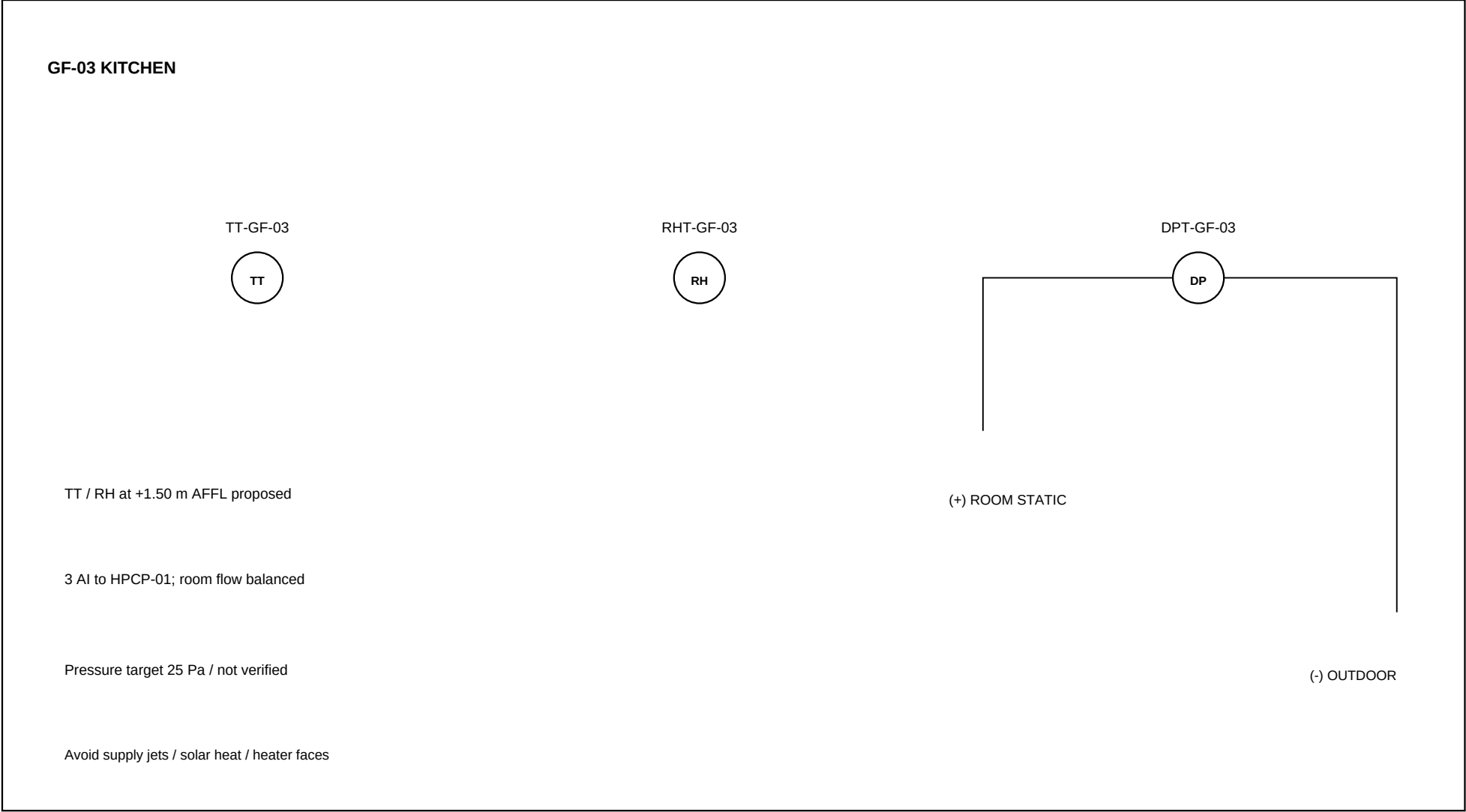
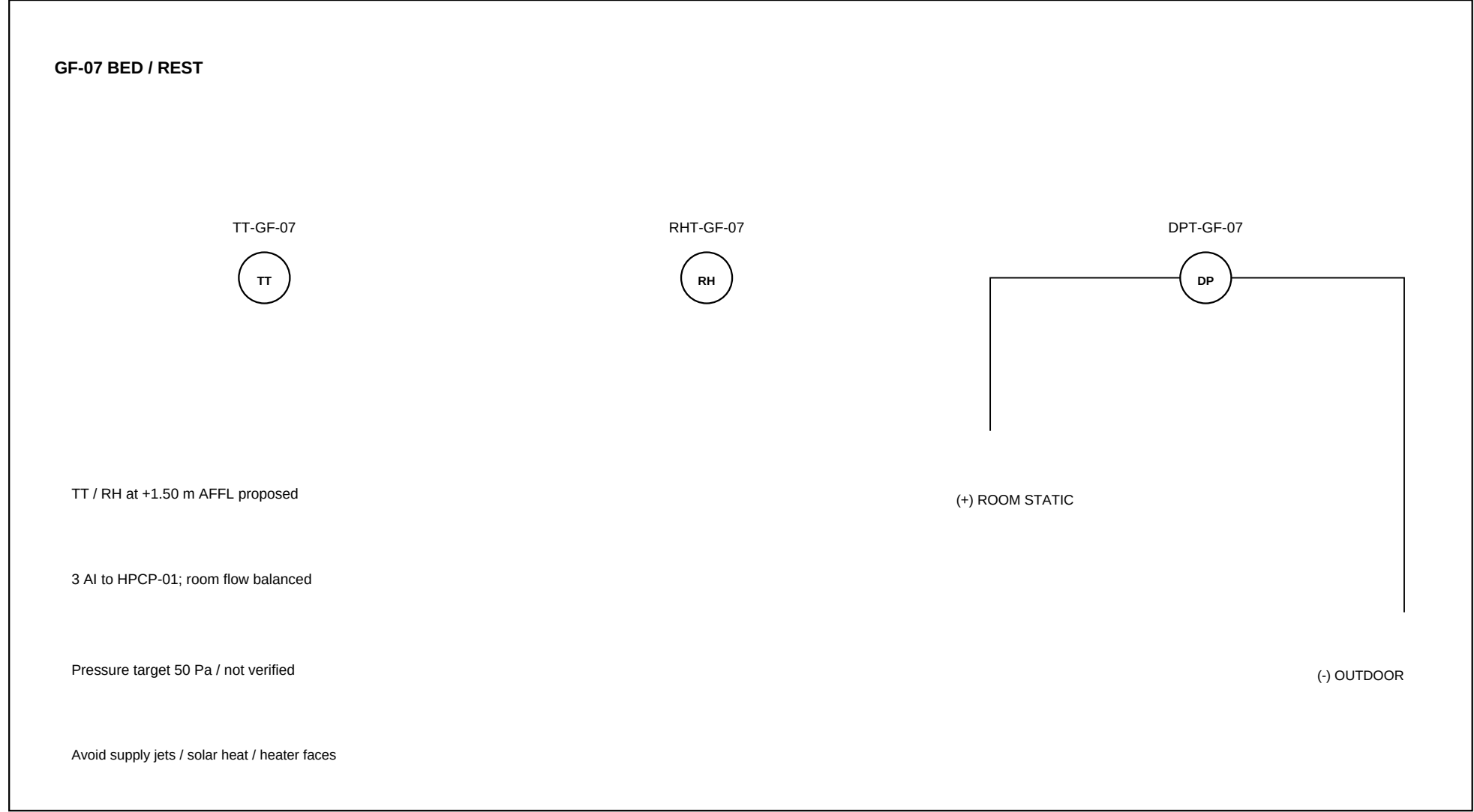
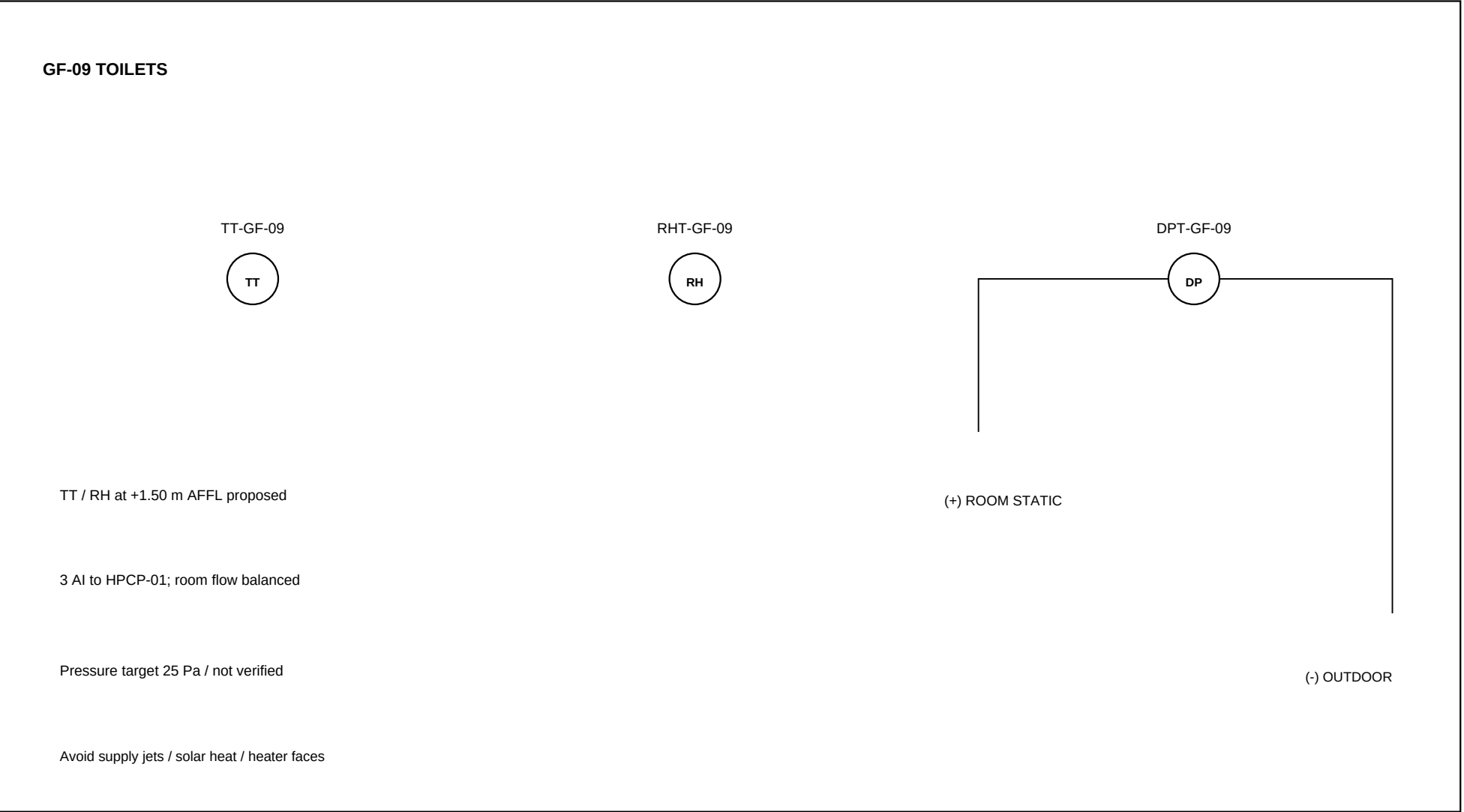
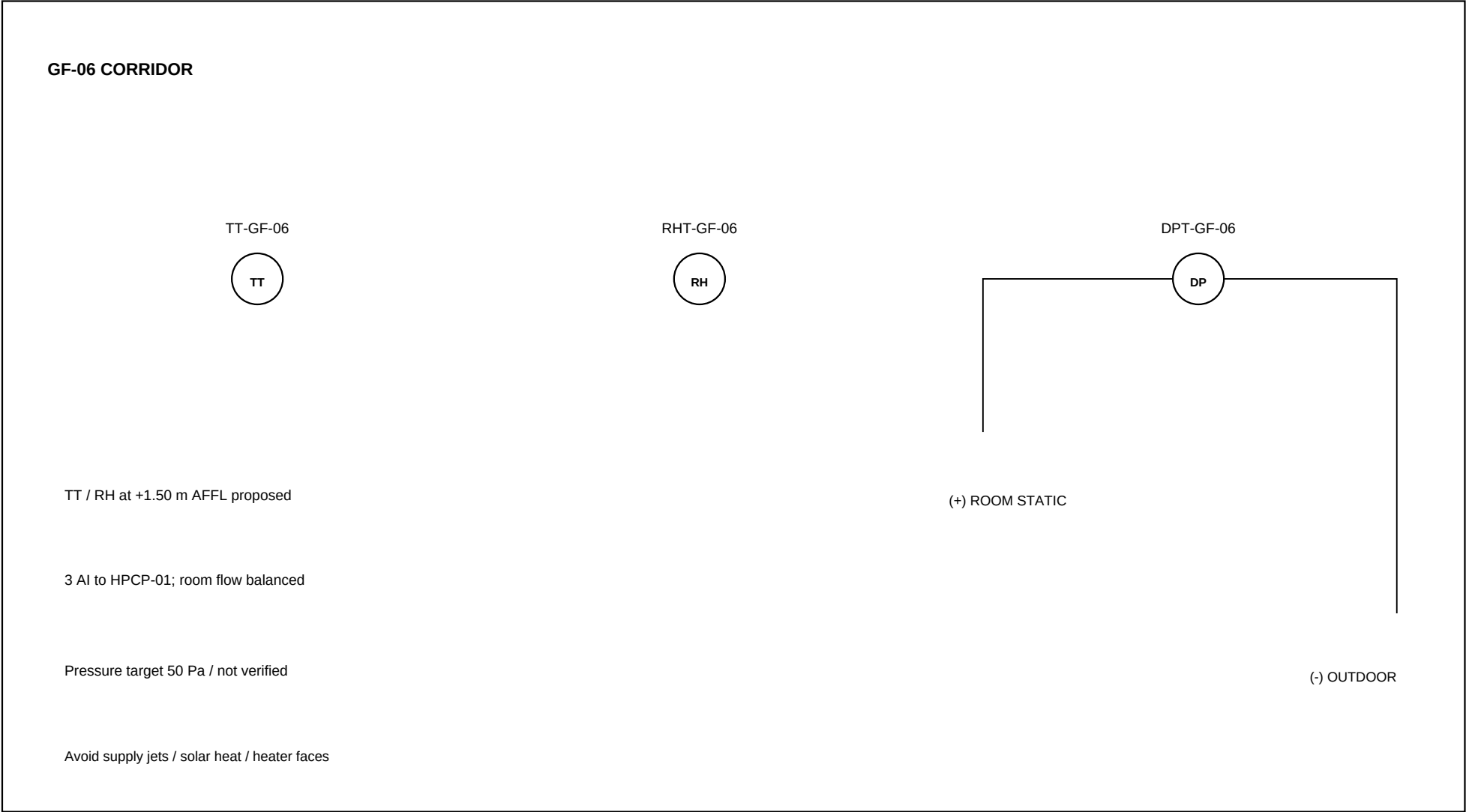
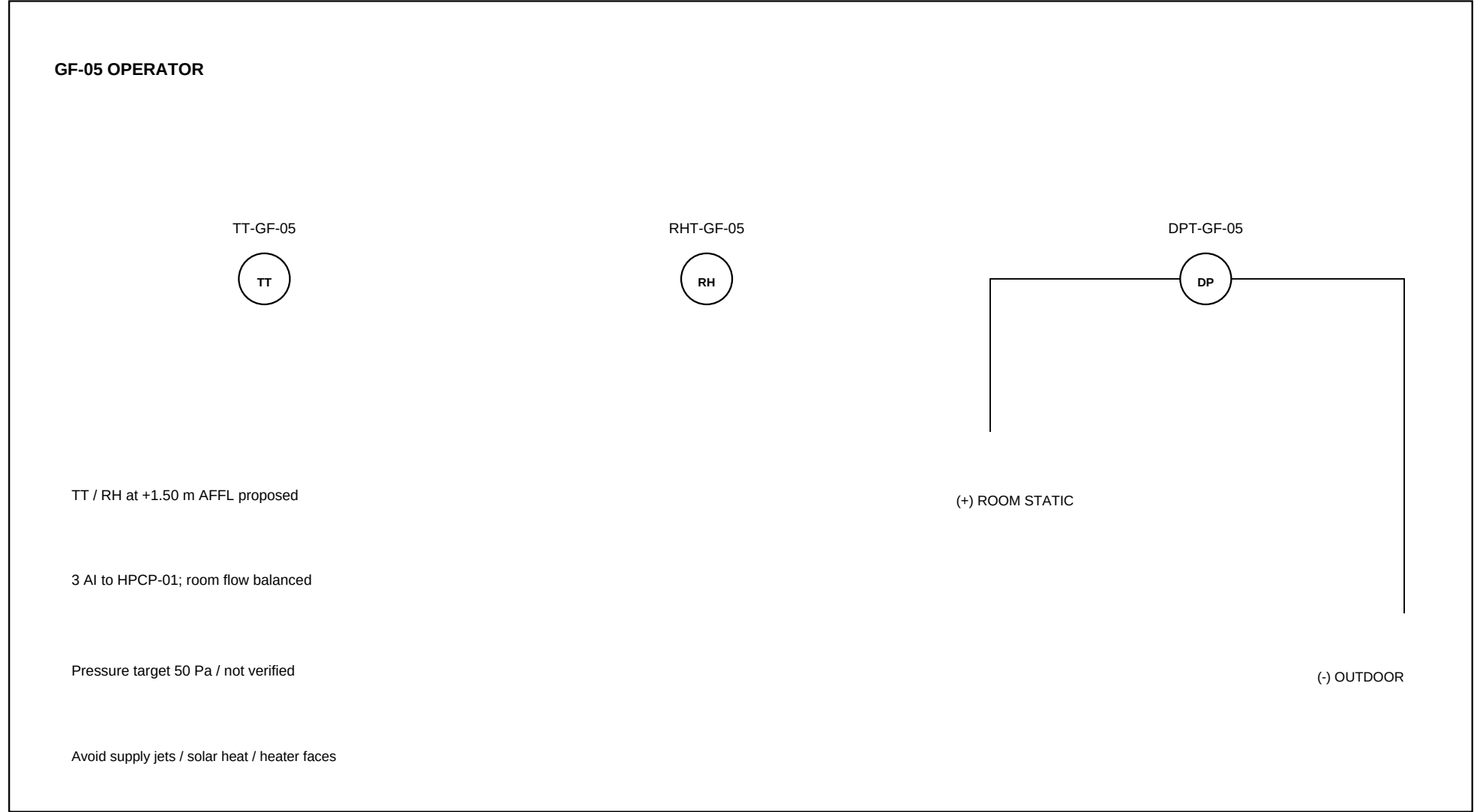
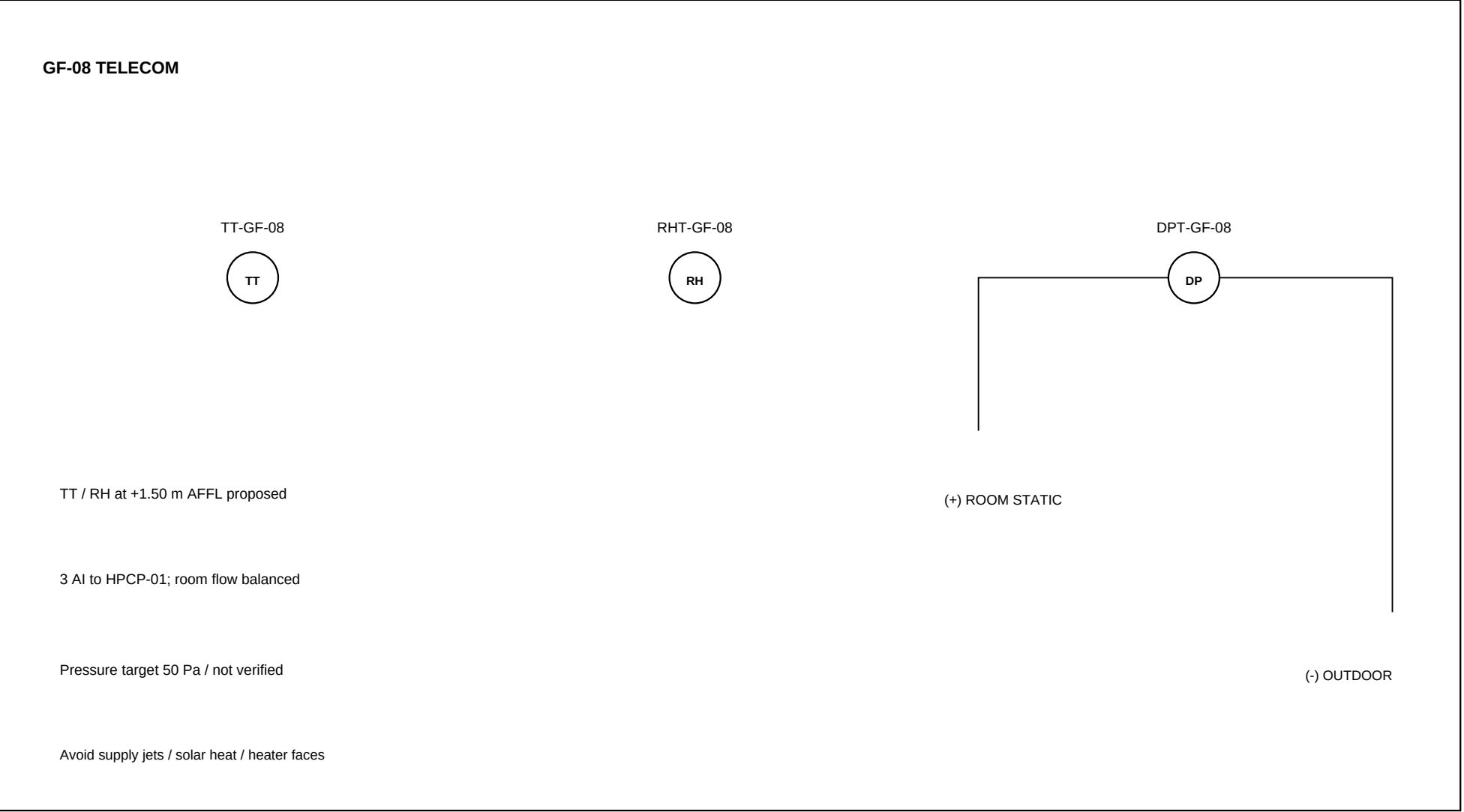
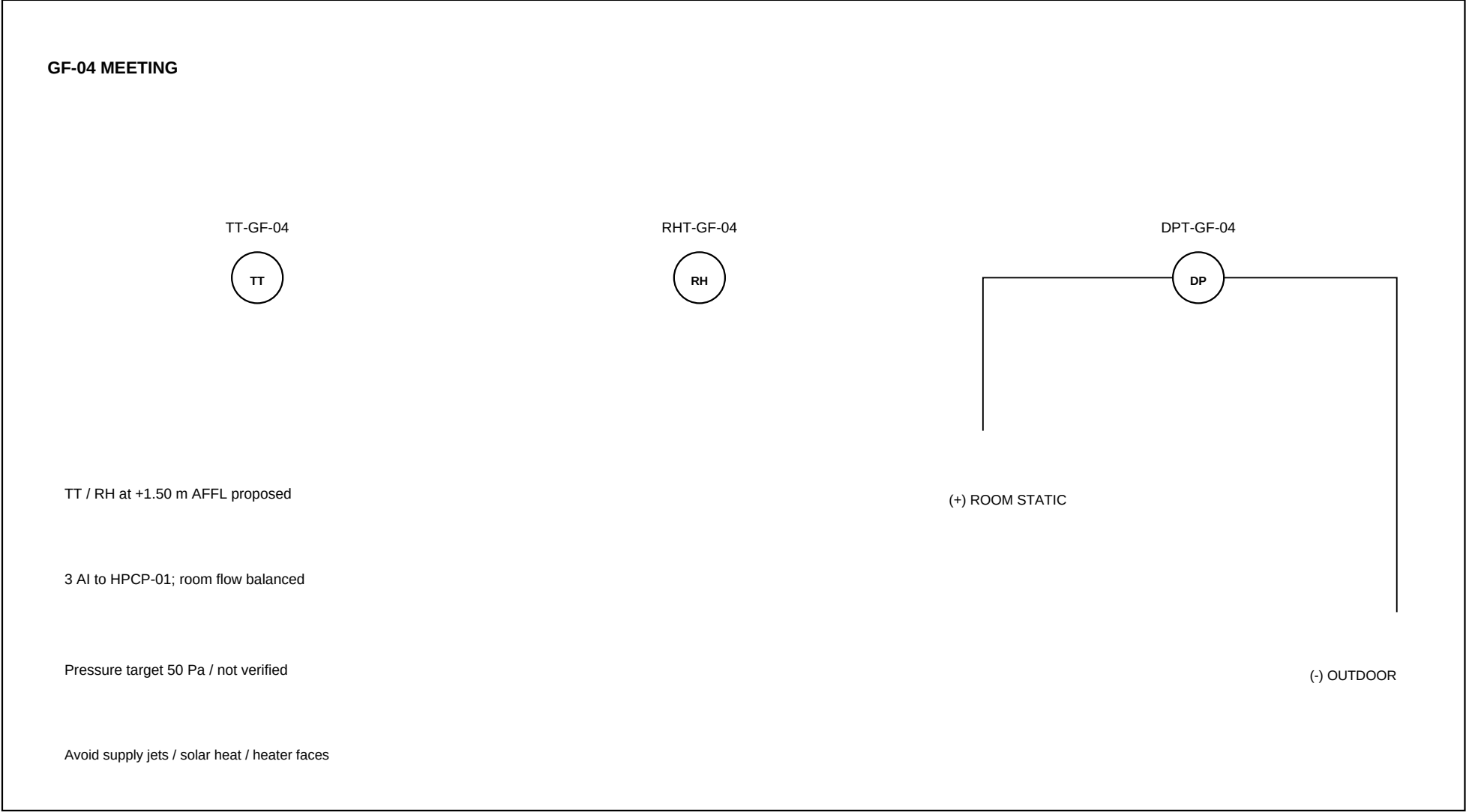
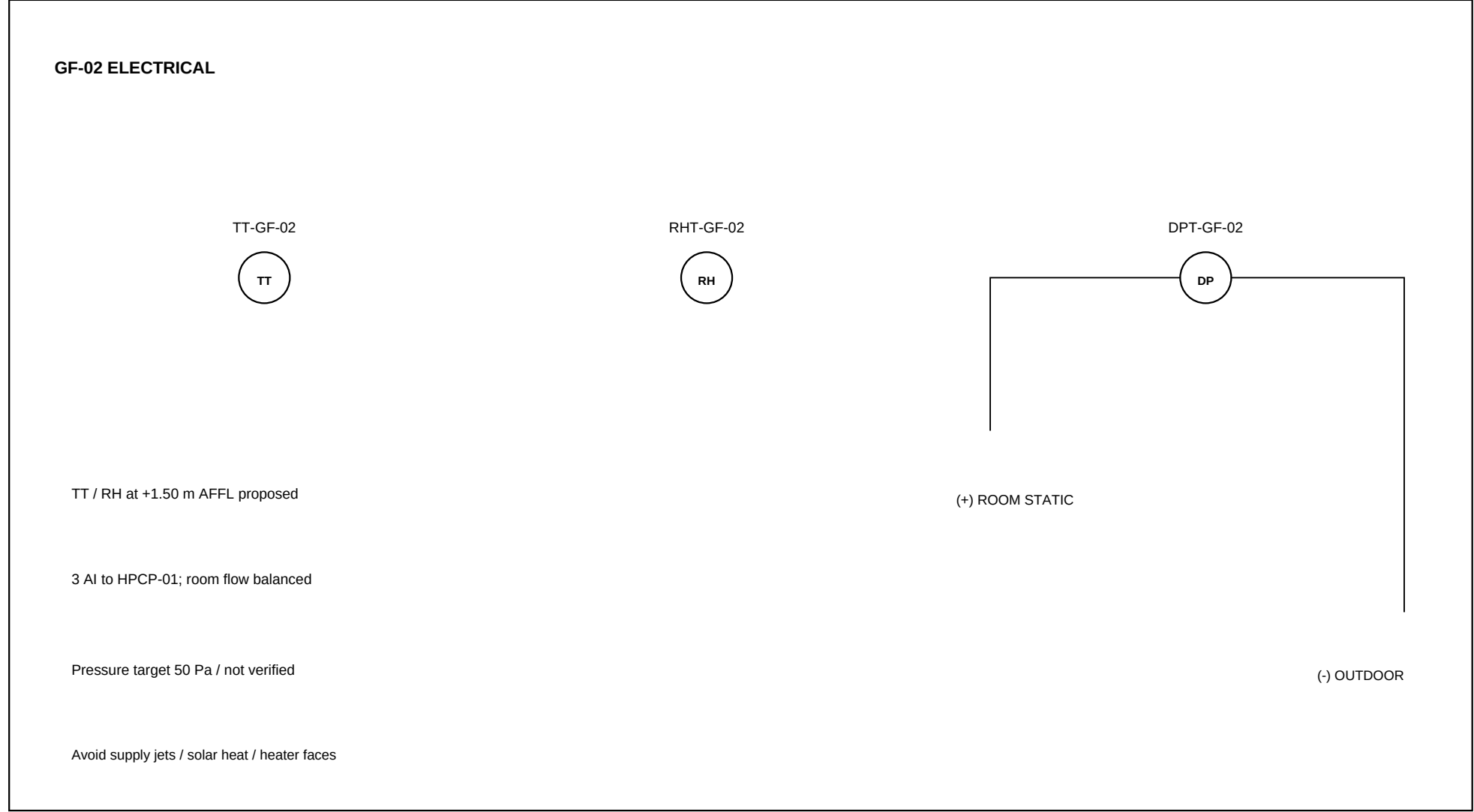


OPTIONAL COMMON-MAIN STATIC CONTROL / NEW PROPOSAL



22 new DX protection / monitoring candidates are separate from the 85 retained air-side instruments. Factory compressor / condenser / evaporator control governs. The shown circuit explains DX operation; receiver, accumulator, refrigerant, valve, oil return, relief and electrical arrangements follow the selected OEM circuit.

No field refrigerant network to the rooms is proposed. Cooling capacity, refrigerant pressure-temperature ranges and compressor quantity/staging are TBC. Do not modify factory safety wiring from this diagram.



Room pressure uses a sheltered outdoor reference, equal datums and static pickups. RCD control is a proposed pressure trim; TT/RHT monitor the room and do not create independent DX zones. Final pressure, comfort control, ranges, setpoints and tubing are pending.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

ROOM TEMPERATURE, HUMIDITY AND PRESSURE INSTRUMENTS

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-006

SCALE: NTS AT A1

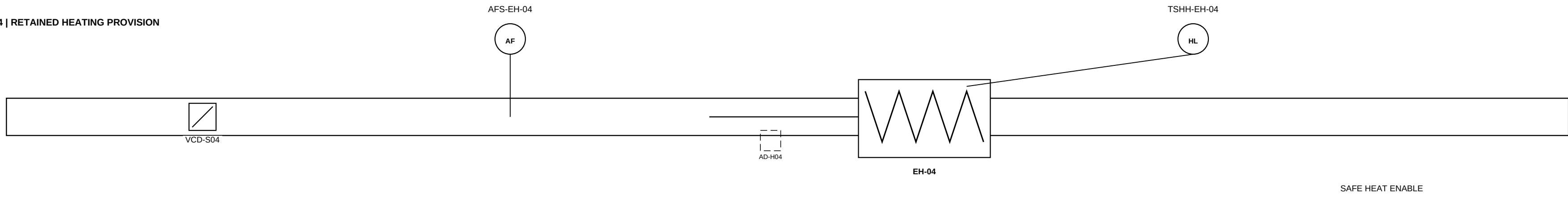
METHOD REVISION 01

08 OCT 2026

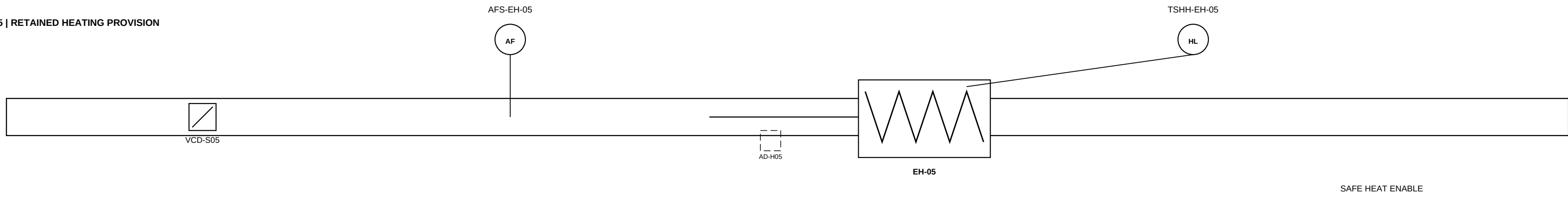
SHEET 6 OF 14

MAIN DUCT AND RETURN - HEATER PERMISSIVES, ACCESS AND HARDWIRED SAFETY

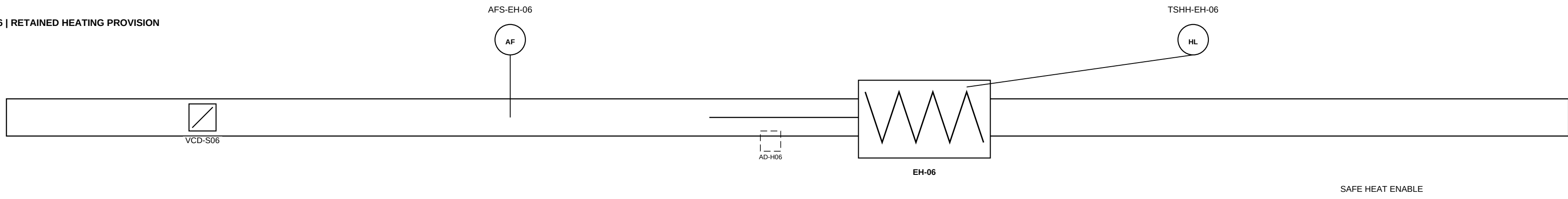
EH-04 | RETAINED HEATING PROVISION



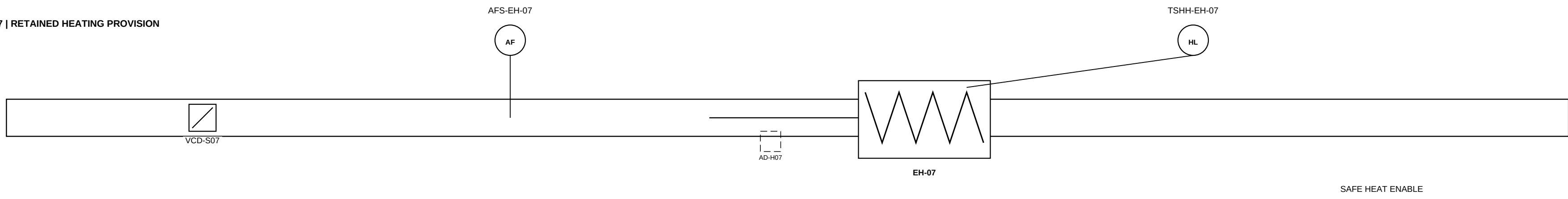
EH-05 | RETAINED HEATING PROVISION



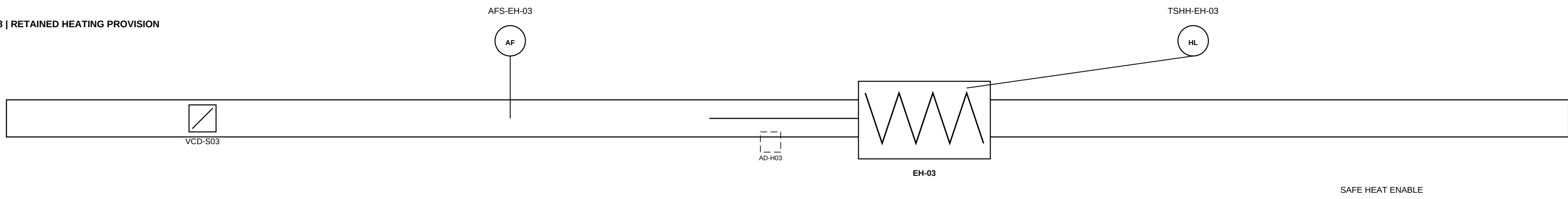
EH-06 | RETAINED HEATING PROVISION



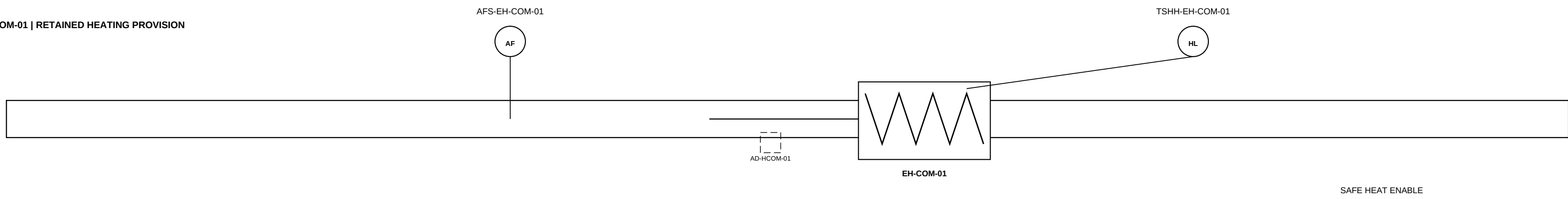
EH-07 | RETAINED HEATING PROVISION



EH-03 | RETAINED HEATING PROVISION

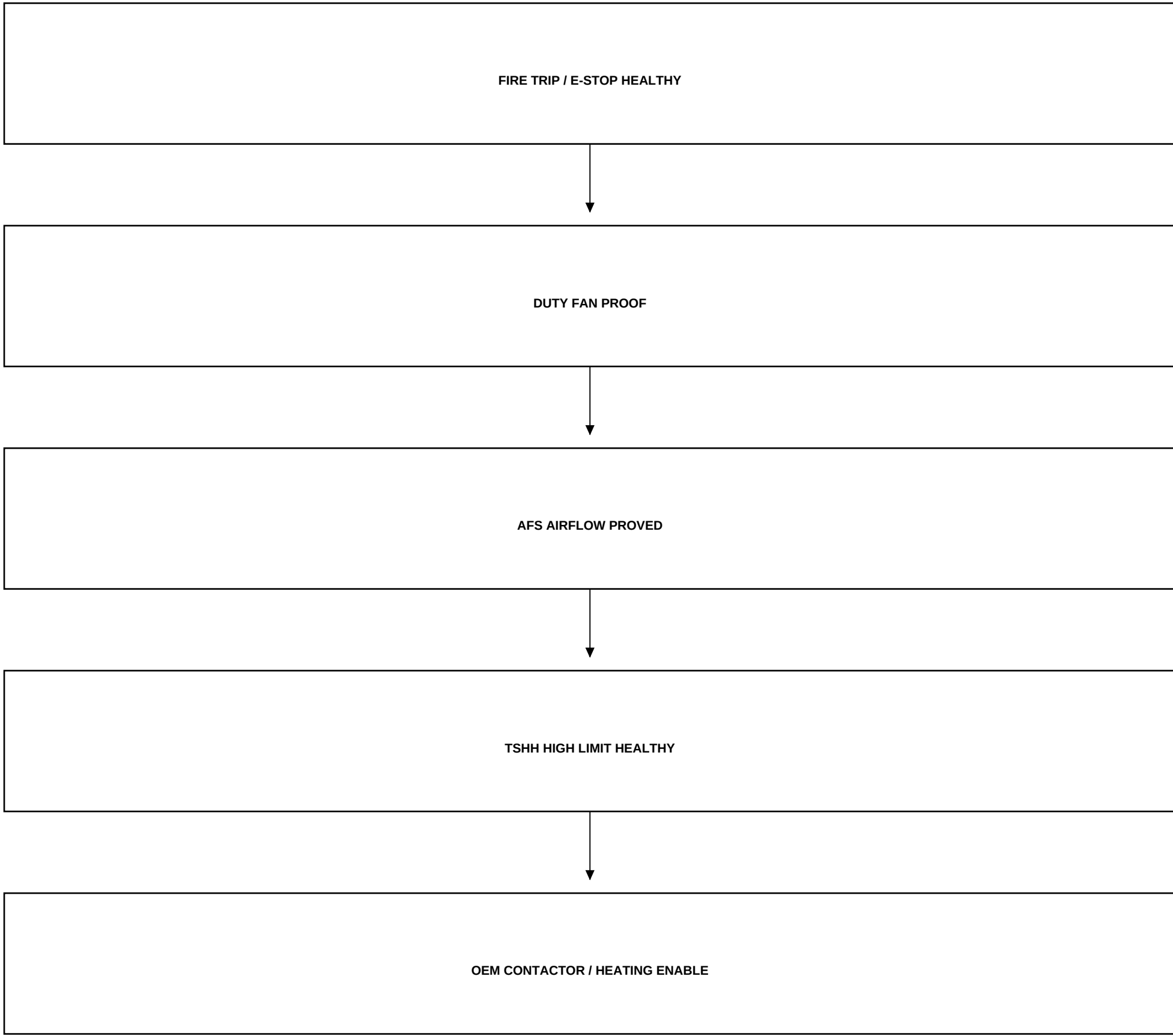


EH-COM-01 | RETAINED HEATING PROVISION



Six AFS and six independent TSHH provisions are retained. Heater dimensions, power, thermal clearances, airflow proof technology and contact reset are OEM selections. Dampers/access symbols are functional; MFD quantity/rating follows the fire strategy.

OEM SAFETY CHAIN / INTERFACE



Loss of airflow or high-limit trip inhibits heat through the independent OEM circuit. HPCP heating demand cannot bypass either safety. Contacts and final electrical terminals follow the selected heater manufacturer.

DX provides cooling. The retained room/common electric heaters and conditional HUM-01 serve distinct heating/humidity duties. Omitting them from a cooling-only operation requires confirmation of the project thermal/humidity requirements; no automatic deletion is made.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

HEATER PERMISSIVES, ACCESS AND HARDWIRED SAFETY

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-007

SCALE: NTS AT A1

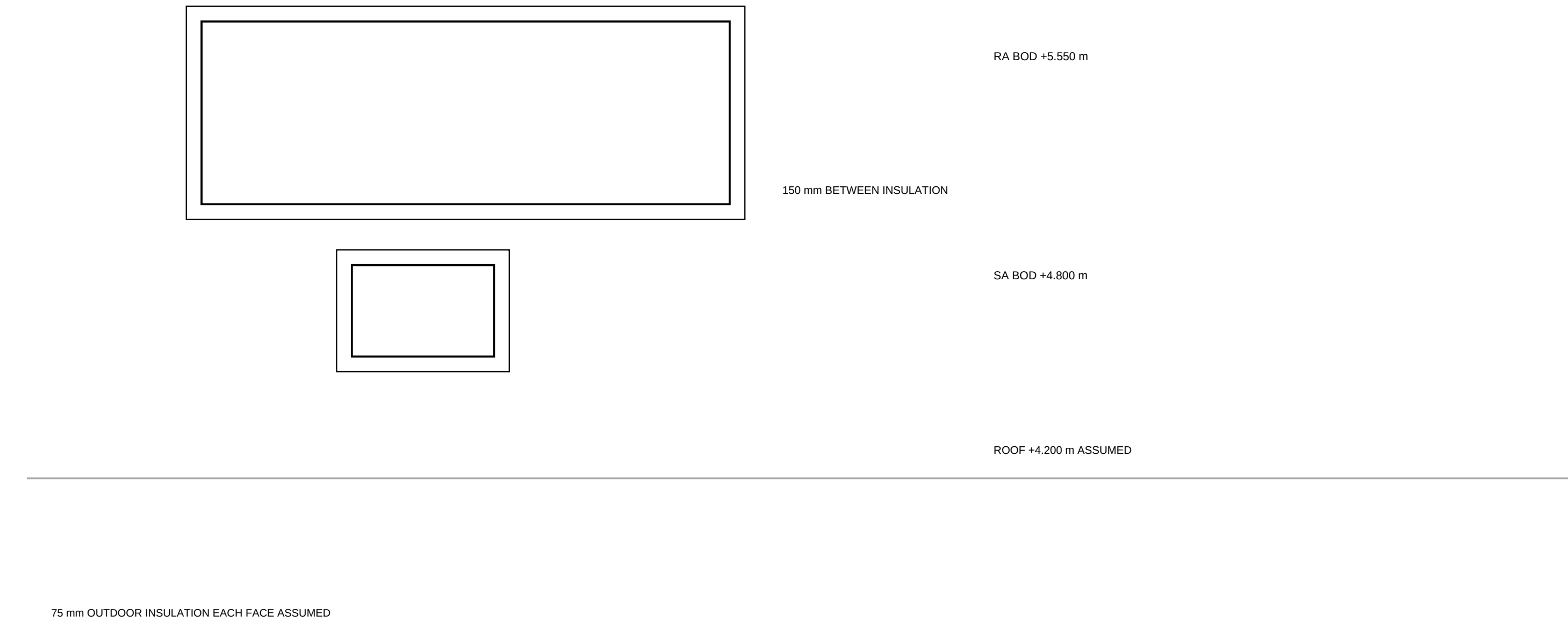
METHOD REVISION 01

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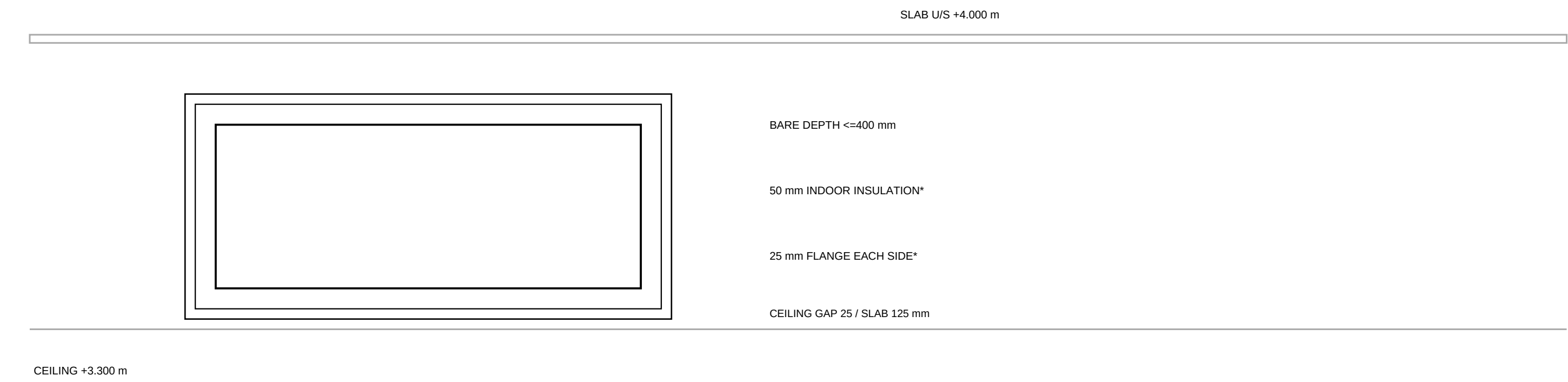
SHEET 7 OF 14

MAIN DUCT AND RETURN - ROOF CROSSING, ROOM DROPS AND CEILING ENVELOPES

A ROOF BRANCH CROSSING / RCX-01 | VERTICAL 1:20



B ROOM DUCT + FLANGES | SECTION 1:10



15 ROOM DROPS: 9 SUPPLY + 6 RETURN / PROPOSED LEVELS

Riser	Room	Clear W x H	Indoor BOD	Roof BOD	Rise m*
RS-SA02	02	500x350	+3.400	4.800	1.4
RS-RA02	02	650x400	+3.400	5.550	2.15
RS-SA04	04	350x400	+3.400	4.800	1.4
RS-RA04	04	500x350	+3.400	5.550	2.15
RS-SA08	08	600x400	+3.400	4.800	1.4
RS-RA08	08	850x400	+3.400	5.550	2.15
RS-SA05	05	600x350	+3.400	4.800	1.4
RS-RA05	05	700x400	+3.400	5.550	2.15
RS-SA06	06	450x300	+3.400	4.800	1.4
RS-RA06	06	500x400	+3.400	5.550	2.15
RS-SA09	09	200x150	+3.400	4.800	1.4
RS-SA07	07	300x250	+3.400	4.800	1.4
RS-RA07	07	350x250	+3.400	5.550	2.15
RS-SA03	03	300x200	+3.400	4.800	1.4
RS-SA01	01	300x200	+3.400	4.800	1.4

CONNECTION / INSTALLATION HOLDS

Roof SA main 600x1300 and RA 700x900 stay outdoors. Only local room ducts descend into the 700 mm ceiling void. Roof-header plan separation and registered crossing levels are shown on MDR-012.

Each RS tag matches its room branch. Two fitted elbows / roof transition, curb / weather seal / firestop and branch access require shop development. Roof +4.200 m assumes a 200 mm slab above the confirmed +4.000 m underside.

* Proposed envelope / rise only. Actual roof structure, beam downstands, insulation specification, actuators, heaters, supports and plenums require OEM/site coordination.

User selected roof common mains with room drops. A shows an actual registered roof branch crossing, not stacked indoor common mains; B uses the retained deepest local-room envelope. Insulation and flange dimensions are assumptions.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

ROOF CROSSING, ROOM DROPS AND CEILING ENVELOPES

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-008

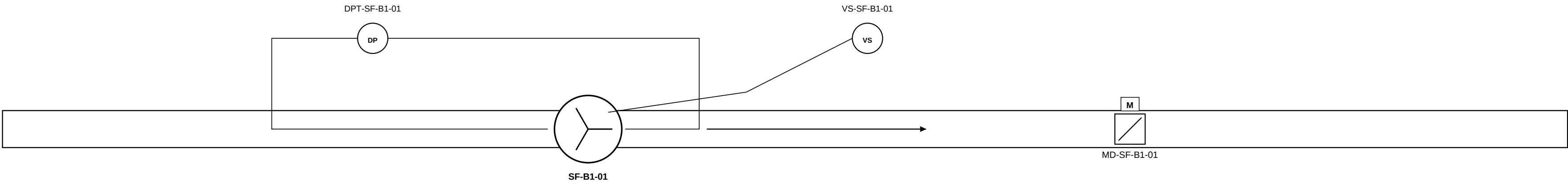
SCALE: NTS AT A1

METHOD REVISION 01

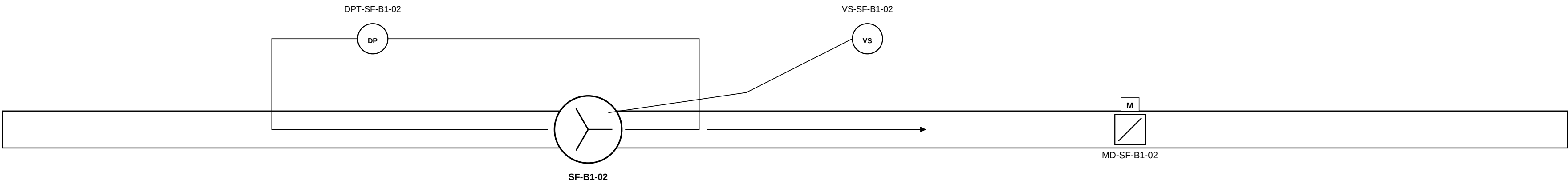
08 OCT 2026

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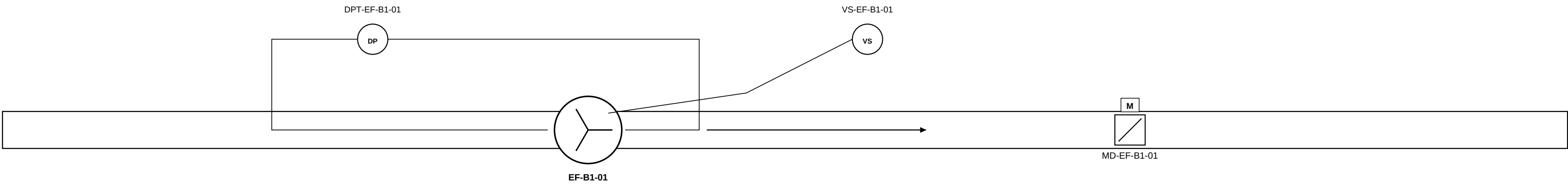
MAIN DUCT AND RETURN - BASEMENT, HPCP, ENERGY METERING AND CONTROL SEQUENCE



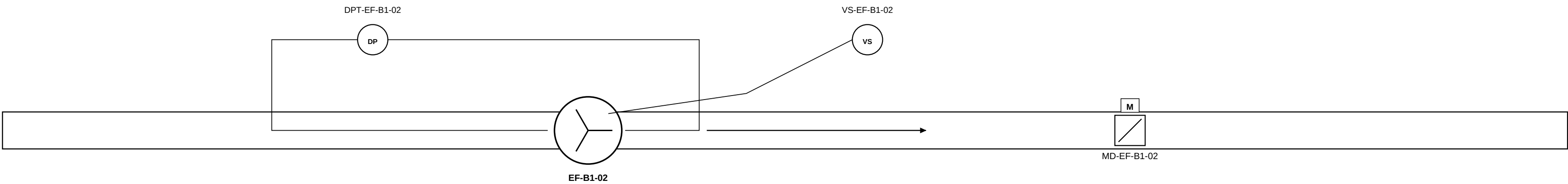
2 x 100% duty / standby, basement flow / heat / volume TBC



2 x 100% duty / standby, basement flow / heat / volume TBC



2 x 100% duty / standby, basement flow / heat / volume TBC



2 x 100% duty / standby, basement flow / heat / volume TBC

Retained I/O	AI	AO	DI	DO	COMM
275 baseline points	65	22	141	44	3

Basement remains an independent ventilation system; its flows/ESP/heat removal are not included in the 3697.80 U/s room supply. New DX proposals are separate from the unchanged 275-row I/O baseline; final channel allocation and protocols pending.

BASEMENT / POWER INSTRUMENTS



HPCP-01 / HMI / BMS SEQUENCE

Start: healthy fire/ESD and OEM safety; select the duty PAU pair; open/prove RA, OA and SA dampers before enabling selected fans/compressors.

Prove fan operation; verify common airflow; enable DX through the OEM controller. Heating requires separate hardwired AFS/TSHH healthy permissives.

Rotate duty pair; failed unit stops / isolates, standby opens/proves then starts. Avoid reverse airflow through an idle unit.

Confirmed fire stops all fans/heaters and latches ESD; manual reset after clearance. Unit compressor response and damper fire positions per approved OEM/fire design.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

BASEMENT, HPCP, ENERGY METERING AND CONTROL SEQUENCE

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-009

SCALE: NTS AT A1

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MAIN DUCT AND RETURN - COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

Retained tag	Parameter	Graphic ref / I/O
TT-GF-02	Dry bulb temperature	MDR-006 / AI
RHT-GF-02	Relative humidity	MDR-006 / AI
DPT-GF-02	Room pressure vs outdoor	MDR-006 / AI
TT-GF-04	Dry bulb temperature	MDR-006 / AI
RHT-GF-04	Relative humidity	MDR-006 / AI
DPT-GF-04	Room pressure vs outdoor	MDR-006 / AI
TSHH-EH-04	Heater high temperature limit	MDR-007 / DI
TT-GF-08	Dry bulb temperature	MDR-006 / AI
RHT-GF-08	Relative humidity	MDR-006 / AI
DPT-GF-08	Room pressure vs outdoor	MDR-006 / AI
TT-GF-05	Dry bulb temperature	MDR-006 / AI
RHT-GF-05	Relative humidity	MDR-006 / AI
DPT-GF-05	Room pressure vs outdoor	MDR-006 / AI
TSHH-EH-05	Heater high temperature limit	MDR-007 / DI
TT-GF-06	Dry bulb temperature	MDR-006 / AI
RHT-GF-06	Relative humidity	MDR-006 / AI
DPT-GF-06	Room pressure vs outdoor	MDR-006 / AI
TSHH-EH-06	Heater high temperature limit	MDR-007 / DI
TT-GF-09	Dry bulb temperature	MDR-006 / AI
RHT-GF-09	Relative humidity	MDR-006 / AI
DPT-GF-09	Room pressure vs outdoor	MDR-006 / AI
TT-GF-07	Dry bulb temperature	MDR-006 / AI
RHT-GF-07	Relative humidity	MDR-006 / AI
DPT-GF-07	Room pressure vs outdoor	MDR-006 / AI
TSHH-EH-07	Heater high temperature limit	MDR-007 / DI
TT-GF-03	Dry bulb temperature	MDR-006 / AI
RHT-GF-03	Relative humidity	MDR-006 / AI
DPT-GF-03	Room pressure vs outdoor	MDR-006 / AI
TSHH-EH-03	Heater high temperature limit	MDR-007 / DI
TT-GF-01	Dry bulb temperature	MDR-006 / AI
RHT-GF-01	Relative humidity	MDR-006 / AI
DPT-GF-01	Room pressure vs outdoor	MDR-006 / AI
TT-SA-01	Supply air temperature	MDR-004 / AI
TT-RA-01	Return air temperature	MDR-004 / AI
RHT-RA-01	Return relative humidity	MDR-004 / AI
FIT-SA-01	Common supply airflow	MDR-004 / AI
FIT-RA-01	Common return airflow	MDR-004 / AI
PDT-OA-01	Outdoor intake filter differential pressure	MDR-004 / AI
PDI-OA-01	Local intake filter differential pressure gauge	MDR-004 / LOCAL
TSHH-EH-COM-01	Common heater high-limit	MDR-007 / DI
PDT-F1-A	Filter differential pressure	MDR-004 / AI
PDI-F1-A	Local filter differential pressure gauge	MDR-004 / LOCAL
PDT-F2-A	Filter differential pressure	MDR-004 / AI

Retained tag	Parameter	Graphic ref / I/O
PDI-F2-A	Local filter differential pressure gauge	MDR-004 / LOCAL
DPT-SF-A	Evaporator fan differential pressure	MDR-004 / AI
VS-PAU-A	RMS vibration velocity	MDR-004 / AI
PDT-F1-B	Filter differential pressure	MDR-004 / AI
PDI-F1-B	Local filter differential pressure gauge	MDR-004 / LOCAL
PDT-F2-B	Filter differential pressure	MDR-004 / AI
PDI-F2-B	Local filter differential pressure gauge	MDR-004 / LOCAL
DPT-SF-B	Evaporator fan differential pressure	MDR-004 / AI
VS-PAU-B	RMS vibration velocity	MDR-004 / AI
PDT-F1-C	Filter differential pressure	MDR-004 / AI
PDI-F1-C	Local filter differential pressure gauge	MDR-004 / LOCAL
PDT-F2-C	Filter differential pressure	MDR-004 / AI
PDI-F2-C	Local filter differential pressure gauge	MDR-004 / LOCAL
DPT-SF-C	Evaporator fan differential pressure	MDR-004 / AI
VS-PAU-C	RMS vibration velocity	MDR-004 / AI
DPT-EF-03-01	Extract fan pressure proof	MDR-003 / AI
VS-EF-03-01	RMS vibration velocity	MDR-003 / AI
DPT-EF-03-02	Extract fan pressure proof	MDR-003 / AI
VS-EF-03-02	RMS vibration velocity	MDR-003 / AI
DPT-EF-09-01	Extract fan pressure proof	MDR-003 / AI
VS-EF-09-01	RMS vibration velocity	MDR-003 / AI
DPT-EF-09-02	Extract fan pressure proof	MDR-003 / AI
VS-EF-09-02	RMS vibration velocity	MDR-003 / AI
DPT-EF-01-01	Extract fan pressure proof	MDR-003 / AI
VS-EF-01-01	RMS vibration velocity	MDR-003 / AI
DPT-SF-B1-01	Fan pressure proof	MDR-009 / AI
VS-SF-B1-01	RMS vibration velocity	MDR-009 / AI
DPT-SF-B1-02	Fan pressure proof	MDR-009 / AI
VS-SF-B1-02	RMS vibration velocity	MDR-009 / AI
DPT-EF-B1-01	Fan pressure proof	MDR-009 / AI
VS-EF-B1-01	RMS vibration velocity	MDR-009 / AI
DPT-EF-B1-02	Fan pressure proof	MDR-009 / AI
VS-EF-B1-02	RMS vibration velocity	MDR-009 / AI
TT-B1-01	Cable basement air temperature	MDR-009 / AI
DPT-B1-01	Basement pressure vs outdoor	MDR-009 / AI
EM-HPCP-01	Energy metering: V, I, kW, kWh, PF, Hz, THD for VFD	MDR-009 / COMM
AFS-EH-04	Airflow proving switch for heater safety permissive	MDR-007 / DI
AFS-EH-05	Airflow proving switch for heater safety permissive	MDR-007 / DI
AFS-EH-06	Airflow proving switch for heater safety permissive	MDR-007 / DI
AFS-EH-07	Airflow proving switch for heater safety permissive	MDR-007 / DI
AFS-EH-03	Airflow proving switch for heater safety permissive	MDR-007 / DI
AFS-EH-COM-01	Airflow proving switch for heater safety permissive	MDR-007 / DI

85 retained instruments are graphically referenced on MDR-003 through MDR-009. 22 new DX candidates are on MDR-005 and in Additional_DX_proposals. Original instrument requirements and I/O are unchanged; final OEM selection / safety settings remain open.

ADMINISTRATION HVAC / DX PAU OPTION

COMMON SUPPLY MAIN + COMMON RETURN MAIN

FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION

DRAWING TITLE

COMPLETE INSTRUMENT INDEX AND DRAWING REFERENCES

ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING

ADM-HVAC-MDR-010

SCALE: NTS AT A1

METHOD REVISION 01

08 OCT 2026

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MAIN DUCT AND RETURN - COMMON HEADER FLOW ACCOUNTING AND ENGINEERING BASIS

Main	Takeoff GF	Before L/s	Branch L/s	After L/s	Clear W x H	m/s	Straight Pa/m
SA-M01	02	3697.80	664.80	3033.00	600x1300	4.741	0.260
SA-M02	04	3033.00	449.20	2583.80	600x1300	3.888	0.180
SA-M03	08	2583.80	884.60	1699.20	600x1300	3.313	0.134
SA-M04	05	1699.20	777.70	921.50	600x1300	2.178	0.062
SA-M05	06	921.50	418.20	503.30	600x1300	1.181	0.021
SA-M06	09	503.30	46.20	457.10	600x1300	0.645	0.007
SA-M07	07	457.10	168.20	288.90	600x1300	0.586	0.006
SA-M08	03	288.90	147.60	141.30	600x1300	0.370	0.003
SA-M09	01	141.30	141.30	0.00	600x1300	0.181	0.001
RA-M01	02	3006.68	605.46	2401.22	700x900	4.773	0.277
RA-M02	04	2401.22	389.86	2011.35	700x900	3.811	0.183
RA-M03	08	2011.35	825.26	1186.09	700x900	3.193	0.132
RA-M04	05	1186.09	718.36	467.73	700x900	1.883	0.050
RA-M05	06	467.73	358.86	108.86	700x900	0.742	0.009
RA-M06	07	108.86	108.86	0.00	700x900	0.173	0.001

CALCULATION / FABRICATION BASIS

Retained roof spines: SA600x1300, RA700x900. This sequential room flow accounting explains the method; actual grouped roof route / branch dimensions and duties are authoritative in Roof_routes_Rev12 and MDR-012. Colebrook/Darcy-Weisbach, rho1.2, mu1.81e-5, galvanized roughness0.09 mm. These are preliminary design choices, not selected acoustic performance.

At each room SA takeoff, remaining flow decreases. On RA the sequence is listed away from the PAU; actual airflow collects toward the PAU. The scaled roof uses two grouped SA branch columns, so these sequential steps are not a fabrication main schedule. No toilet/kitchen/CAG return enters this header.

Develop each reducer with L >= max(300, max((delta W)/(delta H))^2 tan15deg)). Flared three-leg boots and open throats; selected turning vanes/radii and device face-to-face lengths govern shop drawings.

THERMAL / PRESSURE / SELECTION HOLD

SA 3697.80 minus RA 3006.68 requires makeup >= 691.12 L/s. OA reference420.70 is short by 270.42 L/s. This is a mass-balance lower bound, not an approved outdoor-air duty.

No complete fan ESP is calculated for this method: surveyed common-main/riser lengths, nozzle losses, branch layout, fitted K, terminals, coils/dirty filters, intake and exhaust discharge are unselected. The Rev12 allowance budget applies only to its measured proposed room-drop routes, excludes PAU bank/nozzle/OA details and does not select a fan for this method package.

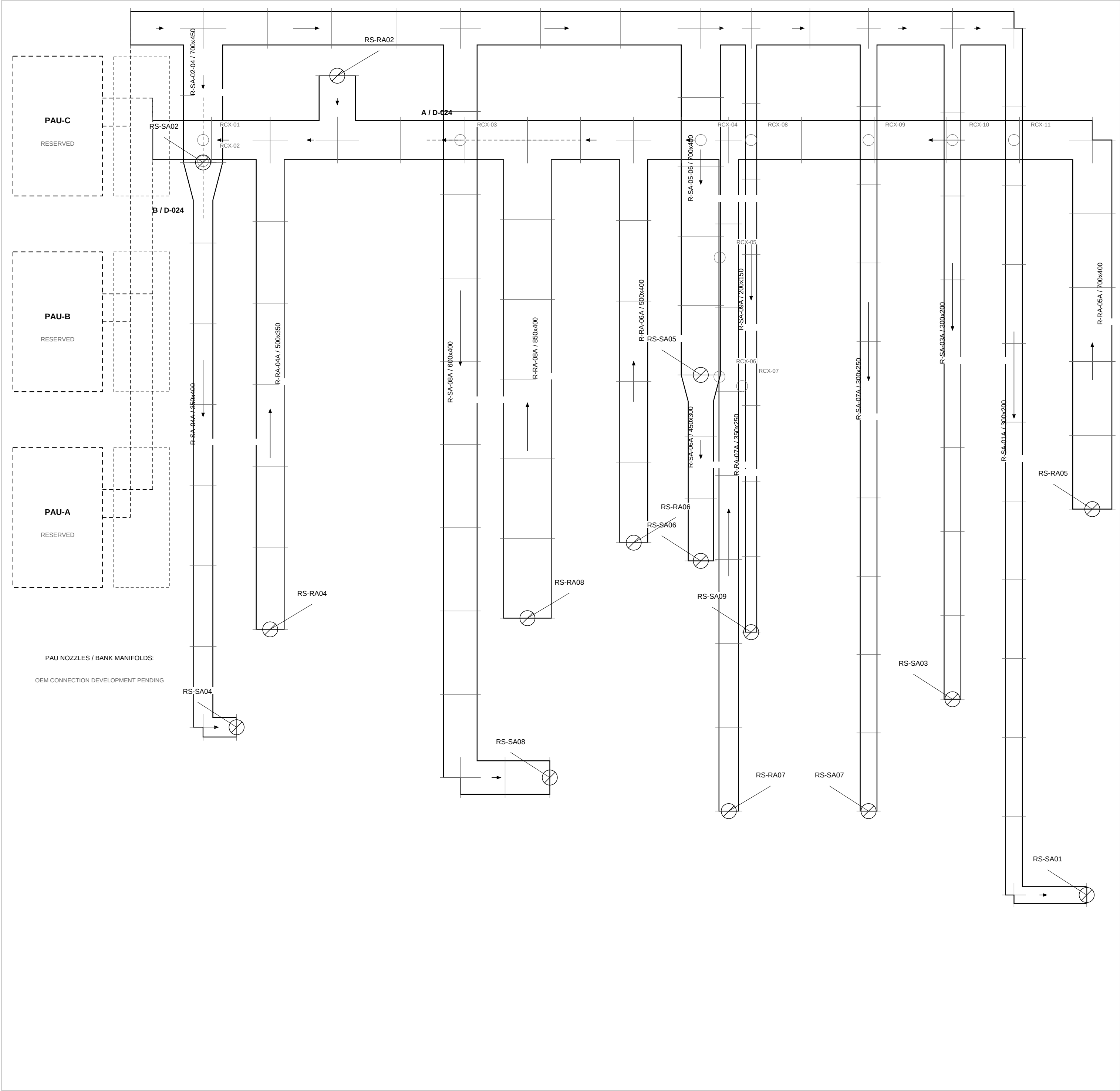
DX coil total/sensible/latent capacity and refrigerant must be selected from revised room/OA loads and design outdoor conditions. One common supply condition alone does not prove independent room comfort; verify loads, reheat/humidity requirements and zoning.

Review index: 001 whole method; 002 all terminals; 003 extracts; 004 PAU/air instruments; 005 DX circuit; 006 room instruments; 007 heater safety; 008 sections/risers; 009 controls/basement; 010 all 85 instruments; 011 functional flow steps; 012 scaled roof plan; 013 indoor coordination; 014 room instrument locations.

ADMINISTRATION HVAC / DX PAU OPTION	DRAWING TITLE	ADM-HVAC-MDR-011	METHOD REVISION 01
COMMON SUPPLY MAIN + COMMON RETURN MAIN	COMMON HEADER FLOW ACCOUNTING AND ENGINEERING BASIS		08 OCT 2026
FOR ENGINEERING COORDINATION - NOT FOR CONSTRUCTION	ROUTES / OEM SELECTION / CHECKED AND APPROVED: PENDING	SCALE: NTS AT A1	SHEET 11 OF 14

ADMINISTRATION BUILDING - ROOF DUCT ROUTING PROPOSAL

MDR-012 REFERENCE PLAN / RETAINED REV12 D-023 / MAIN DUCT AND RETURN



Roof levels use provisional roof surface +4.200 m; user underside +4.000 m plus assumed 200 mm slab. Route lengths are measured proposals, not surveyed as-built lengths. All sizes are clear internal mm.

ROOF ROUTES / LEVELS

31 connected route segments feed all 15 room supply / return drops. Coordinates use the supplied floor plan origin. Roof footprint and PAU reservation zones must be surveyed.

GREEN SUPPLY: bare BOD +4.800 m; main 600 x 1300 mm. BLUE RETURN: bare BOD +5.550 m; main 700 x 900 mm. Tall roof mains limit plan width; wind / parapet screening and support stability require structural design.

75 mm outdoor insulation each face assumed. Eleven roof plan overlaps have separated insulation envelopes; minimum calculated separation 150 mm. No upper return riser intersects lower supply insulation in this proposed geometry.

PAU-A / B / C reservations: 1600 x 2500 mm each, 1000 mm east service strip. Dashed nozzle stubs and bank manifolds are schematic interfaces awaiting manufacturer dimensions.

RCX tags: Roof_crossings_Rev12 register. A and B sections: D-024. Curbs and supports: D-030. Native centerline setting-out dimensions: D-032.

Service	Main clear mm	Bare BOD	Insulated top
SA	600 x 1300	+4.800	+6.175
RA	700 x 900	+5.550	+6.525

ACCESS / LOAD HOLD

Service strips are reserved for PAUs only. Walking platforms, safe roof access, guarding, filter pull-out, common heater / conditional humidifier weather enclosure, banks and lifting path require selected equipment layout. Roof proposal is not an approved installation.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOF DUCT ROUTING PROPOSAL

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-023

SCALE: 1:50 AT A1

REVISION

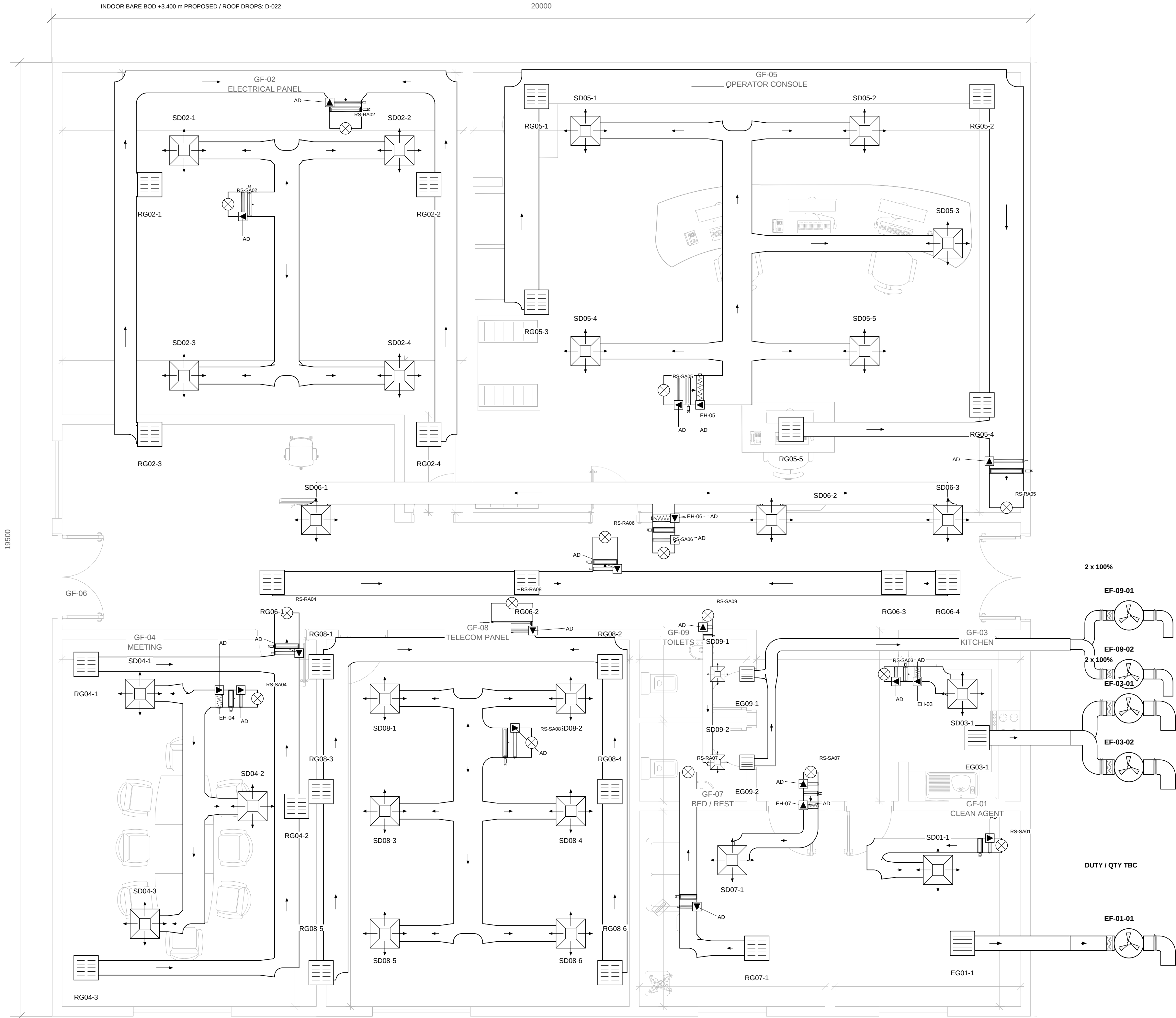
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ADMINISTRATION BUILDING - COORDINATED DUCT PLAN

MDR-013 REFERENCE PLAN / RETAINED REV12 D-001 / MAIN DUCT AND RETURN



ROOM AIRFLOW SCHEDULE

GF	Room	SA L/s	RA L/s*	EA L/s*
02	ELECTRICAL PANEL	664.8	605.5	-
05	OPERATOR CONSOLE	777.7	718.4	-
04	MEETING	449.2	389.9	-
08	TELECOM PANEL	884.6	825.3	-
06	CORRIDOR	418.2	358.9	-
07	BED / REST	168.2	108.9	-
09	TOILETS	46.2	-	64.9
03	KITCHEN	147.6	-	147.6
01	CLEAN AGENT	141.3	-	TBC

DUCT / SYMBOL LEGEND

	SA - supply duct
	RA - return duct
	EA - dedicated extract
	SD - four-way supply diffuser
	RG - ducted return grille
	EG - extract grille
	VCD - balancing damper
	MFD* - conditional fire damper
	RCD - return control damper
	EH - inline electric heater
	AD - access door
	RS - proposed roof supply / return drop

Terminal aliases: SD02-1 = SD-02-01; RG / EG follow the same format. Full tags and quantities are in the registers.

Roof-main OPTION: 15 independent room drops RS-SA / RS-RA. No indoor service crossings. Finished ceiling +3.300 m; proposed bare BOD +3.400 m. Envelope: D-020; roof/drop holds: D-022.

* Return quantities are assumed balances. Extract references and +50 / +25 Pa targets require final airflow and pressure verification.

Units: mm / L/s / m/s. Terminal faces and clear duct widths are shown. Duct levels, selected fitting development and manufacturer clearances remain pending. No shared return from kitchen, toilet or clean-agent room.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

COORDINATED DUCT PLAN

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-001

SCALE: 1:50 AT A1

REVISION

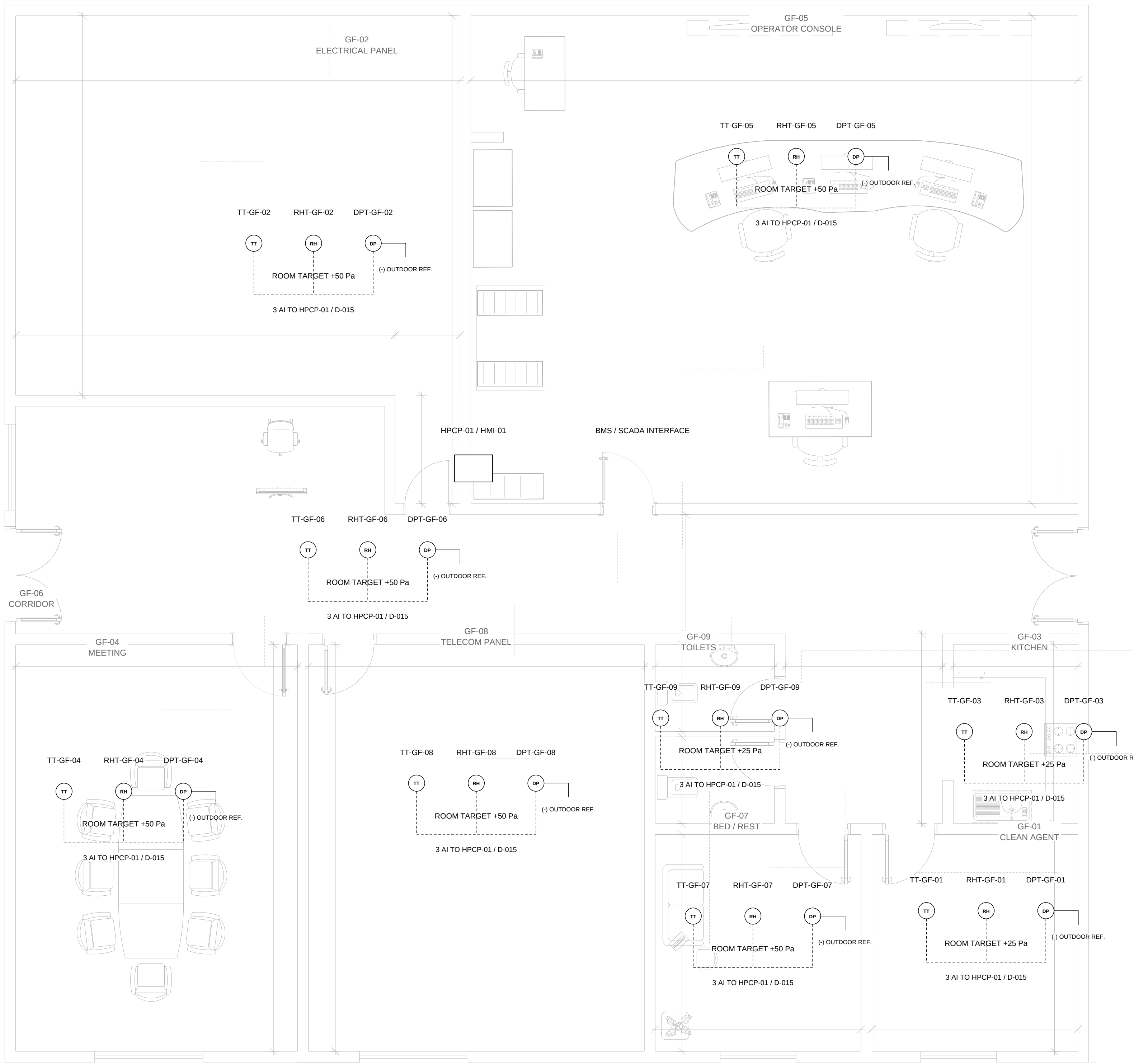
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ADMINISTRATION BUILDING - ROOM INSTRUMENTATION AND CONTROL LOCATIONS

MDR-014 REFERENCE PLAN / RETAINED REV12 D-007 / MAIN DUCT AND RETURN



Room instrument locations are proposed. Plant/filter/fan instruments and heater AFS / TSHH protection: D-012. Final ranges, reference probes, mounting and routes require I&C coordination; operating sequence: S-004.

ROOM FIELD INSTRUMENTS

GF	Temperature	Humidity	Pressure	Pa
02	TT-GF-02	RHT-GF-02	DPT-GF-02	50
05	TT-GF-05	RHT-GF-05	DPT-GF-05	50
04	TT-GF-04	RHT-GF-04	DPT-GF-04	50
08	TT-GF-08	RHT-GF-08	DPT-GF-08	50
06	TT-GF-06	RHT-GF-06	DPT-GF-06	50
07	TT-GF-07	RHT-GF-07	DPT-GF-07	50
09	TT-GF-09	RHT-GF-09	DPT-GF-09	25
03	TT-GF-03	RHT-GF-03	DPT-GF-03	25
01	TT-GF-01	RHT-GF-01	DPT-GF-01	25

MAIN DUCT / PLANT MONITORING

TT-SA-01 / TT-RA-01: main duct temperature

RHT-RA-01: main return humidity

FIT-SA-01 / FIT-RA-01: common airflow

Each PAU filter: PDT remote + PDI local

DPT-SF-A/B/C: PAU fan pressure proof

VS: PAU and ventilation-fan vibration

EM: feeders >7.5 kW, quantity to confirm

Signals report to HPCP-01 dedicated PLC and HMI-01. BMS / SCADA provides the specified monitoring, initial start and emergency-stop interfaces. Protocol, alarm limits, I/O module allocation and actuator fail positions remain pending.

DPT room references are outdoors, not the adjacent room. DPT-GF-06 is the corridor / building reference. Additional room DPTs and CAG humidity sensing are marked as added provisions in the register.

TAQA TRANSMISSION

N-19082-RO-012 | PUMPING STATION STANDARDIZATION

ADMINISTRATION HVAC

DESIGN BASIS: C-H-SS-001 Rev0 / HAP REFERENCE

DRAWING TITLE

ROOM INSTRUMENTATION AND CONTROL LOCATIONS

FOR COORDINATION | CHECKED / APPROVED: PENDING

DRAWING NUMBER

ADM-HVAC-D-007

SCALE: 1:50 AT A1

REVISION

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